

Multimodal fusion with LLM content via hierarchical progressive transformer for explainable fake news detection

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ABSTRACT

Detecting fake news on social media has become a significant challenge due to the rapid dissemination of misleading information. Current works often focus on combining limited modalities, lacking deep and progressive integration of diverse features. Apart from textual and image features, there remains a challenge in incorporating other features while also ensuring the explainability of the detection results. To address these problems, we propose a novel Explainable Fake News Detection method that integrates Multimodal Fusion with Large Language Model-generated content (LLM-MFEFND). This model utilizes Large Language Models (LLMs) to generate background knowledge and comments to integrate the LLM's interpretation of news. To extract deep and cross-modal knowledge, we propose Hierarchical Progressive Transformer (HPT) to integrate images, texts, aligned multimodal features, background information, and user comments deeply and progressively. Furthermore, we design Multimodal Local Interpretable Model-agnostic Explanations (MLIME) method to provide explanations for the detection results. Through extensive experiments on datasets such as WeiBo21 (6,010 news items) and FineFake (15,140 news items), we demonstrate that the LLM-MFEFND model using DeepSeek outperforms existing methods. Specifically, our method achieves an accuracy of 94.5% on WeiBo21 and 81.1% on FineFake, representing improvements of at least 1.7% and 2.2% respectively over existing approaches. By combining LLM-driven context enrichment with robust multimodal fusion and explainability, our framework advances the fight against misinformation. The code is available at <https://github.com/li1608335419/LLM-MFEFND>.

1. Introduction

Social media is the primary channel for users to acquire knowledge, such as Twitter and Facebook in English-speaking countries, and platforms like Weibo, Rednote, and Toutiao in Chinese-speaking regions. For instance, Weibo attracts 598 million monthly visitors who browse for information. However, fake news has become a pressing societal concern, owing largely to its widespread dissemination through social media platforms (Zhou & Zafarani, 2018). Generative AI advancements further exacerbate this issue. For example, a large amount of fake news can be generated by LLMs like ChatGPT, GPT-4, and DeepSeek (Neha & Bhati, 2025). While human fact-checking professionals are capable of accurately verifying the authenticity of news, their efforts struggle to keep pace with the sheer volume of online content. It is evident that if all fake news are to be verified by fact-checking professionals, it would

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Fig. 1. Fake news examples: (a) Drinking boiled garlic water will cure COVID-19. (b) A picture shows a massive tsunami headed for a crowded beach. (c) The storm wall in real life in Paris. This is always how I've envisioned it.

cause delays in public awareness and consume substantial manpower. Therefore, automatic fake news detection on social platforms is receiving increasing attention from researchers.

Most previous studies predominantly utilize text-based analysis to detect fake news. However, if those who spread fake news alter their writing style, the model's accuracy may decrease. Visual elements play a crucial role in fake news by leveraging multimedia like images and videos to attract and mislead readers, facilitating rapid dissemination. In light of recent developments where fake content integrates audio and video, incorporating analysis of auxiliary elements is crucial to enhance the model's precision (Athira et al., 2023). The research by Jing et al. (2023) shows that tweets with images are much more likely to be retweeted than those without images. Fig. 1 presents three examples of multimodal fake news, each containing both an image and text. In Fig. 1(a), the image in the post appears normal, but the textual content is misleading. In contrast, Fig. 1(b) features normal text, while the image is clearly fabricated. Fig. 1(c) exemplifies a scenario where both the image and the text are disseminating false information. As demonstrated by the above examples, multimodal messages yield better fake news detection results.

Most multimodal fusion methods focus on two or three features (primarily image and text), but research on the fusion of more than three features remains relatively scarce (Comito et al., 2023). Features from different modalities contain redundant, complementary, and relevant semantic information (Chen et al., 2022; Liu et al., 2023; Wu et al., 2023). When dealing with three or more features, it is necessary to focus on progressively deeper fusion and higher-order complementary information preserving during the feature integration process. Moreover, most news texts lack contextual background, making it difficult to accurately identify misinformation. These limitations directly worsen the challenges of fake news detection. Beyond data-level issues, deep learning-based methods suffer from transparency challenges due to the black-box nature of neural networks (Liu et al., 2023; Ma et al., 2023). The explainability of these fake news detection models remains largely obscure, casting doubt on their reliability. Athira et al. (2023) mentioned that researchers have already fused visual and textual information for detecting fake news, but the explanation in this regard has not been addressed. Consequently, more and more researchers focus on the importance of ensuring the reliability and explainability of these systems (Wu et al., 2021b).

In light of the aforementioned analysis, existing fake news detection still faces the following challenges: (1) The limitations posed by sparse semantic representation and incomplete contextual understanding; (2) Difficulty in effectively fusing and integrating more than three features, due to the complex interplay of complementarity, redundancy, and semantic correlations among them; (3) Lack of explainability for detection results, particularly in multimodal fusion scenarios, limiting transparency and trust in model decisions.

To mitigate the above limitations, we propose a novel Explainable Fake News Detection method that integrates Multimodal Fusion with Large Language Model-generated content (LLM-MFEFND). Our method has three components: knowledge extraction, knowledge fusion, and explanation module. The knowledge extraction phase addresses incomplete contextual understanding by integrating multimodal inputs with augmented features, including: 1) background knowledge generated by LLMs, 2) message comments enriched with LLMs' judgments and interpretations, and 3) aligned multimodal features derived from text-image fusion. In knowledge fusion, to achieve deep and progressive integration of data from more than three features, we design the Hierarchical Progressive Transformer (HPT) module for iterative fusion of multimodal data. In explanation module, we introduce the MLIME approach for explainability analysis of tweet texts, images, and the content generated by LLMs to provide a post-hoc explanation (Athira et al., 2023). Furthermore, we use t-SNE to visualize the effect of feature fusion.

In summary, this study aims to leverage LLMs to generate useful features and explore a progressive deep fusion strategy for combining these augmented features with multimodal data, thereby extracting important deep representations as well as cross-modal knowledge between texts and images, while also providing explainability analysis for the detection results under multimodal fusion scenarios. This work fills a gap in previous research.

The contributions in this paper can be summarized as follows:

(1) We propose LLM-MFEFND, a novel multimodal fake news detection model with various modalities and LLM-generated contents that can fully utilize different features from multiple perspectives. (2) We introduce the HPT module to capture the deep fused information across multi-level feature interactions, supporting integration of at least three distinct features. (3) We construct a rigorously expert-verified background and comment collection from LLM for our task on both Chinese and English dataset, and we have made the dataset publicly available. (4) We develop MLIME, a multimodal explanation method that explains the detection results through various features. Comprehensive experimental results on WeiBo21 and FineFake datasets verify the effectiveness of LLM-MFEFND.

The subsequent sections of this work are organized as follows. Section 2 critically surveys prior research on fake news detection, analyzing both the capabilities and shortcomings of current methodologies. Section 3 presents the proposed methodology LLM-MFEFND. Implementation details and obtained results are presented in Section 4, which includes an in-depth assessment of our model's capabilities relative to existing advanced methods, along with comprehensive analyses on parameters, robustness, efficiency, ablation, and explainability. Section 5 discusses the theoretical and practical implications of our model and outlines future works. Section 6 summarizes the challenges and essential findings of the study.

2. Related works

Fake news is defined as intentionally fabricated and disseminated false information with the purpose of misleading readers and making them doubt verifiable facts, and it can be confirmed by another source (Zhang & Ghorbani, 2020). Fake news has become increasingly diverse and continues to evolve, leading existing methods to gradually transition from unimodal to multimodal approaches. Additionally, with the development of LLMs, some research has started to explore the integration of these technologies into the task of fake news detection. This section provides a detailed overview of the development of fake news detection techniques, primarily focusing on multimodal detection, LLM-based detection, and explainability analysis (see Table A.9).

2.1. Multimodal fake news detection

Researchers have recently devoted heightened attention to fake news detection utilizing multimodal techniques, with numerous research outcomes emerging. Gao et al. (2024) employed Knowledge Graph Embedding (KGE) to incorporate knowledge into language and visual embedding model, generating a generic representation for texts and images. Gólo et al. (2023) utilized One-class Learning (OCL) to learn representations from multimodal information, including news content, topic and linguistic information. Tong et al. (2024) proposed a novel multi-domain detection framework by fusing images and texts to construct domain-specific and cross-domain knowledge extraction. Ying et al. (2023) proposed an improved MMoE network, combining different feature refinement, including images semantic, texts and image patterns. By jointly modeling knowledge enhancement and adversarial domain adaptation, combined with multimodal encoding and label reweighting, KEAN (Zhou et al., 2024) addressed the issues of covariate shift and label shift in fake news detection. To model the complex relationship between different modalities, attention mechanism was utilized in some research, such as MCAN (Multimodal Co-Attention Networks) (Wu et al., 2021b), EMIF (Evidence-aware Multi-source Information Fusion) (Dong et al., 2024), and ETMA (Efficient Transformer-based Multilevel Attention framework) (Yadav et al., 2024).

Besides, due to the widespread popularity of short videos, detecting fake news on short video platforms is expected to be a future research trend. Some researchers utilized a visual LLM model (Bu et al., 2024) and multi-view debiasing framework (Zeng et al., 2024) on short video platforms, with the combination of comments, videos, images, audios and news content. However, if multiple modalities are input simultaneously into a fusion model (such as MLP, variational encoder, etc.), it may generate excessive redundant information and result in the loss of cross-modal interactions. In our research, we aim to deeply and progressively fuse text and image data while considering the cross-modal attention relationships.

2.2. LLM-based fake news detection

LLMs have demonstrated strong performance across various classification and reasoning tasks in recent studies. Despite requiring substantial training time on the latest information, many scholars confirm the significant potential of LLMs in fake news detection. Wang et al. (2024a) proposed a defense-based explainable fake news detection framework equipped with LLM model through evidence

Table 1
Key notations.

Notation	Description
$X = (I, T)$	A text-image pair comprising textual component T and visual component I
r_r	LLM-generated background
r_c	LLM-generated comments
ϕ_i	Representation of image I using ViT
ϕ_t	Representation of text T using BERT
R_r	Representation of background r_r using BERT
R_c	Representation of comments r_c using BERT
f_i	Visual (image) feature
f_t	Textual feature
f_{t-i}	Aligned multimodal feature
$f_{r \rightarrow x}$	News-background interaction feature
$f_{c \rightarrow x}$	News-comments interaction feature
F_{fused}	Fused information fed into the classifier

extraction, providing high-quality justifications by inferring reasons from two competing parties. Li et al. (2023) presented SELF-CHECKER, a modular framework utilizing LLMs to enable efficient fact verification with limited training requirements. Hu et al. (2024) found that LLMs were good at offering insightful multi-perspective rationales for small language models to detect fake news despite suboptimal performance in veracity judgment. Li et al. (2024) presented FactAgent, which employs LLMs for detecting fake news by implementing a structured workflow without requiring model training, thereby dividing the intricate task of verifying news authenticity into several steps. Park et al. (2024) introduced AdStyle, an adversarial style augmentation technique, aimed at training a resilient fake news detector capable of withstanding diverse style-conversion attacks. The style-conversion attacks are generated by style conversion prompts. Kareem and Abbas (2023) employed fine-tuned LLMs to detect fake news by generating logical reasoning. Zhou et al. (2025) integrated the functions of LLMs and SLMs, employing a two-stage retrieval process to select relevant knowledge, with the method outperforming the pure SLMs approach. Although LLM-based fake news detection has achieved improvements, the growing integration with LLMs carries risks of uncontrollable problems due to the problem of hallucinations and misalignment, limiting their application in fake news detection (Hoy & Koulouri, 2025). To avoid directly using LLMs for fake news detection, this paper primarily utilizes LLMs to generate additional textual information to assist in fake news detection.

2.3. Explainable fake news detection

Certain researchers strive to attain inherent explainability through weight coefficients (attention mechanisms). Lu and Li (2020) utilized tweets' content, retweet sequences, and user profiles, through the attention weights. The model can achieve explainable detection result. Shu et al. (2019) employed sentence-comment co-attention to capture semantic affinity between news content and comments, while simultaneously learning attention weights for sentences and comments. Han et al. (2024a) proposed a co-attention sub-network for the embedding vectors of comments to apprehend top-k explainable check-worthy user comments and sentences.

Other researchers utilize knowledge graphs or external sources to verify news authenticity and provide explanatory insights. Wu et al. (2021a) determined whether the false news by identifying entities and relationships among entities in false news. This method relies on creating a multi-relation knowledge graph based on the existing false news databases to infer new entities and their relationships. Silva et al. (2021a) proposed Propagation2Vec to enhance the explainability of the model by employing an attention mechanism that assigns different weights to nodes and propagation cascades. However, this method requires obtaining the propagation paths of false news, which is challenging in most studies. Chi and Liao (2022) proposed a decision making system to autonomously gain human-level knowledge and engage users via dialogue trees, aiding comprehension of its internal reasoning. Based on the social network graph and news knowledge graph, in the work of Han et al. (2024b), the explainable reasoning module in this method applies self-attention mechanisms to nodes within the social graph and news knowledge graph. Wei et al. (2025) explicitly extracted three types of cross-modal clues: enhancement, complementarity, and conflict, and used key clues as the basis for explaining the detection results.

Existing methods primarily detect fake news without adequately explaining their results to the public, especially in the context of multimodal feature fusion. Most methods that incorporate attention mechanisms are considered explainable. However, attention mechanisms do not provide transparency regarding the underlying principles and reasoning process. The explainability remains weak, and in certain cases, it can be confusing. In our work, we extend existing explainability methods, centering around local optimization theory, and apply them to the multimodal domain. We conduct explainability analysis on various features (such as text, images, and LLM-generated content) to explain the effectiveness of the proposed model.

3. Methodology

Given a new message represented as a text-image pair $X = (I, T)$, the objective of our work is to determine the authenticity of the new message. The key notations used in this work are summarized in Table 1.

As shown in Fig. 2, we develop a novel multimodal fake news detection model named LLM-MFEFND. We first extract single-modal features and multimodal features from multi-views, including the CLIP (Contrastive Language-Image Pretraining) model (Radford

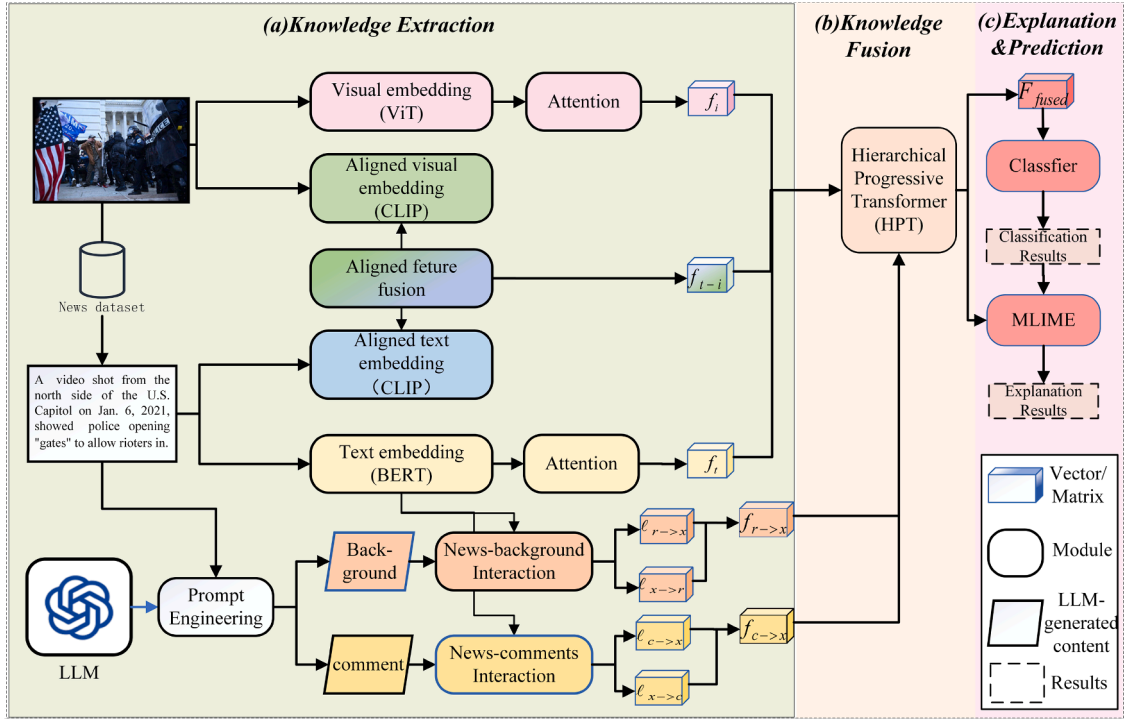


Fig. 2. The workflow of LLM-MFEFND.

et al., 2021), BERT and ViT (Vision Transformer) (He et al., 2022). Then, our method generates external LLM-generated background knowledge and user comments as textual description. Next, the news-background-comments interaction module is designed to enhance the interactions between news and background information, as well as between news and comments. Subsequently, the HPT module fuses information from LLM-generated information and multimodal information. Finally, MLIME is employed to provide explanations for the results.

3.1. Multimodal feature extraction

We conduct multimodal feature extraction from various perspectives. First, we leverage pre-trained models to extract unimodal features from different views. Then, we align the visual and textual features extracted from the CLIP model to derive aligned multimodal features.

3.1.1. Visual features

Given an image I , we scale the image size to 224×224 and obtain the image representation using the ViT module. The model is pre-trained on massive image datasets through a masked autoencoding task, enabling it to effectively capture both global and local features of the image. We denote the image representation as $\phi_i \in \mathbb{R}^{L_i \times d_i}$, where L_i is the number of patches, d_i is the embedding dimension of each patch. Then, we utilize attention with SiLU activation function to capture the importance differences between different regions in the image:

$$S = FC(\text{SiLU}(FC(\phi_i))) \quad (1)$$

$$\alpha = \text{softmax}(S, \text{dim} = 1) \quad (2)$$

$$f_i = \alpha \cdot \phi_i \quad (3)$$

where $FC()$ denotes a full connected layer. In addition, aligned visual embedding f_{c-i} is extracted using CLIP's pre-trained image encoder module.

3.1.2. Text features

Given a text T , we use the pre-trained BERT model to extract the embedding vector of each word in the text T . We denote the embedding vector of the text T as ϕ_t , $\phi_t \in \mathbb{R}^{L_t \times d_t}$, where L_t is the number of tokens, d_t is the dimension of each token. Furthermore, we adopt attention scheme to capture the importance differences between different tokens in the text:

$$f_t = \text{softmax}((FC(\phi_t))^T \cdot \phi_t) \quad (4)$$

In addition, aligned text embedding f_{c-t} is extracted using CLIP's pre-trained text encoder module.

```
PROMPT_TEMPLATE = (
    "You are designed as an auxiliary tool for a fake news detection system. Your main
    function is to provide a brief explanation of the relevant background knowledge and simulate
    five comments from real users based on a given message."
    "Your goal is to output a concise explanation of the background knowledge and user
    comments, separated by commas, which will help people assess the authenticity of the message
    without needing additional background information."
    "Your explanation should be under 300 words. Return the output in JSON format with
    fields named 'background' and 'comment'."
)
```

Fig. 3. The prompt template.

3.1.3. Aligned multimodal features

There exist cross-modal semantic differences between the visual and textual features extracted by ViT and BERT, making it unsuitable to directly fuse these two features. Thus, we employ the CLIP model to obtain aligned textual and visual embedding. We concatenate them to a MLP network to generate the aligned multimodal embedding, which is denoted as f_{t-i} :

$$f_{t-i} = MLP(f_{c-t} \oplus f_{c-i}) \quad (5)$$

3.2. LLM-generated content

3.2.1. Prompt engineering

We evaluate ChatGLM¹, an open-source bilingual Chinese-English conversational LLM released by Zhipu AI. Fine-tuning for specific tasks requires significant time and computational resources, so we adopt the zero-shot prompt learning approach, where the task is learned simply through specific instructions (Li, 2023). Hu et al. (2024) has demonstrated that the current LLMs can serve as a good generator of instructive rationales. The majority of false news on social networks consists of short texts. Expanding the underlying knowledge of these short texts through LLMs could aid in understanding the content of false messages and assessing their authenticity. We design a prompt template to generate extra comments and background knowledge, which is shown in Fig. 3. The prompt template includes function description, goals and requirements, and output format. Since the LLM is not tasked with complex fact-checking, chain-of-thought reasoning, or multi-step decision-making, the generation process remains robust and relatively insensitive to minor variations in prompt phrasing.

Prior to experimentation, all LLM-generated outputs undergo rigorous manual verification by news experts to ensure rationality and relevance. The generated outputs conflicting with image/text modalities are automatically flagged and excluded from the final dataset. In practice, most LLM-generated content demonstrates satisfactory rationality, with only a small portion requiring iterative regeneration and validation until it meets our requirements.

3.2.2. News-background-comments interaction

Given a news message $X = (T, I)$ and its background r_r and comments r_c generated by LLM, we encode the T , r_r and r_c using the BERT to have φ_t , R_r , and R_c . Then, we build news-background and news-comment interaction features via incorporating a bidirectional cross-attention module to promote feature correlations. The bidirectional cross-attention module for news-background correlation feature generation can be shown as follows:

$$\ell_{r \rightarrow x} = AvgPool(CrossAtt(R_r, \varphi_t, \varphi_t)) \quad (6)$$

$$\ell_{x \rightarrow r} = AvgPool(CrossAtt(\varphi_t, R_r, R_r)) \quad (7)$$

$$CrossAtt(R_r, \varphi_t, \varphi_t) = softmax\left(\frac{R'_r \cdot \varphi'_t}{\sqrt{d}}\right)\varphi'_t \quad (8)$$

¹ <https://github.com/THUDM/ChatGLM>

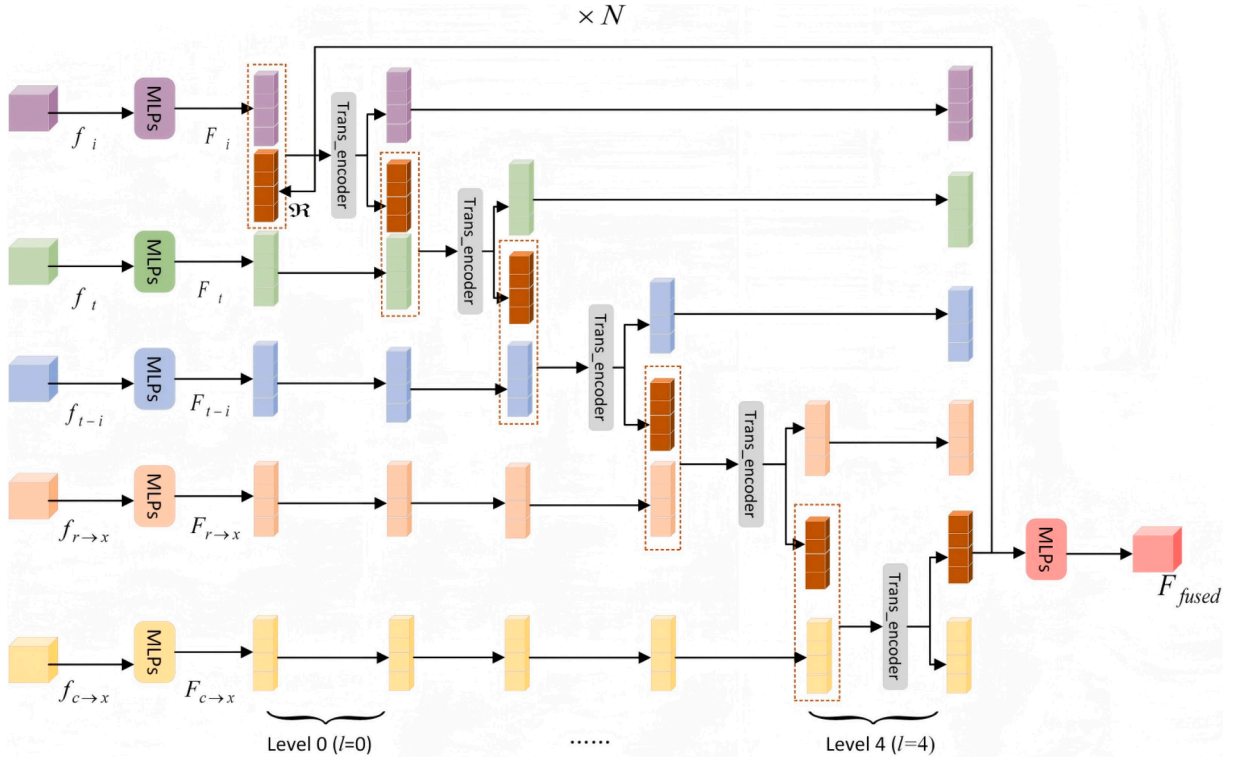


Fig. 4. The structure of Hierarchical Progressive Transformer.

where $R'_r = W_r R_r$, $\varphi'_t = W_t \varphi_t$, d means the dimensionality. $AvgPool()$ refers to performing average pooling on the token embeddings of the cross-attention output. We input $\ell_{x \rightarrow r}$ into a MLP module to obtain a scalar value $w_{x \rightarrow r}$, which is used to re-weight the vector $\ell_{r \rightarrow x}$:

$$f_{r \rightarrow x} = w_{x \rightarrow r} \cdot \ell_{r \rightarrow x} \quad (9)$$

The same operation can be performed to obtain news-comment correlation feature $\ell_{c \rightarrow x}$ and $\ell_{x \rightarrow c}$:

$$\ell_{c \rightarrow x} = AvgPool(CrossAtt(R_c, \varphi_t, \varphi_t)) \quad (10)$$

$$\ell_{x \rightarrow c} = AvgPool(CrossAtt(\varphi_t, R_c, R_c)) \quad (11)$$

$$CrossAtt(R_c, \varphi_t, \varphi_t) = softmax\left(\frac{R'_c \cdot \varphi'_t}{\sqrt{d}}\right) \varphi'_t \quad (12)$$

where $R'_c = W_c R_c$. $\ell_{x \rightarrow c}$ is then input into a MLP module to obtain a scalar value $w_{x \rightarrow c}$ to re-weight $\ell_{c \rightarrow x}$:

$$f_{c \rightarrow x} = w_{x \rightarrow c} \cdot \ell_{c \rightarrow x} \quad (13)$$

3.3. Hierarchical Progressive Transformer

Upon deriving the embeddings of the aforementioned five features, including image (f_i), text (f_t), aligned multimodal fusion (f_{t-i}), news-background interaction ($f_{r \rightarrow x}$), and news-comments interaction ($f_{c \rightarrow x}$), it is critical to design an effective model for their integration. As shown in Fig. 4, unlike simple concatenation $f_i \oplus f_t \oplus f_{t-i} \oplus f_{r \rightarrow x} \oplus f_{c \rightarrow x}$, which merely stacks multimodal features without interaction and often introduces redundant information due to its equal treatment of all modalities, HPT employs a hierarchical attention mechanism to progressively fuse information. After generating feature embeddings via MLP, HPT first builds cross-modal interactions and dependencies through progressive Transformer encoder layers. Then, it iteratively refines these representations through multiple progressive Transformer stages to fully integrate information while extracting complementary features. This architecture enables the model to capture effectively fused representations, a capability quantitatively confirmed in our ablation study (see Fig. 9 in Section 4.4).

Firstly, we generate feature embedding sequences through several MLP networks:

$$F_i = \text{Concat}(\text{MLPs}(f_i)) \quad (14)$$

$$F_t = \text{Concat}(\text{MLPs}(f_t)) \quad (15)$$

$$F_{t-i} = \text{Concat}(\text{MLPs}(f_{t-i})) \quad (16)$$

$$F_{r \rightarrow x} = \text{Concat}(\text{MLPs}(f_{r \rightarrow x})) \quad (17)$$

$$F_{c \rightarrow x} = \text{Concat}(\text{MLPs}(f_{c \rightarrow x})) \quad (18)$$

where $F_i, F_t, F_{t-i}, F_{r \rightarrow x}, F_{c \rightarrow x} \in \mathbb{R}^{m \times q}$ represent the generated embedding sequences, q is the number of MLP networks.

Secondly, we create a sequence combination $\mathcal{R}^l = [R_1, R_2, \dots, R_q]^l \in \mathbb{R}^{m \times q}$ to progressively fuse information from different features, l is the index of the fusion level, $l \in [0, 4]$. The steps of hierarchical progressive fusion can be summarized as follows:

$$\text{Level } 0/l = 0 : [F_i^1, \mathcal{R}_i^1] = \text{Trans_encoder}([F_i^0, \mathcal{R}^0]) \quad (19)$$

$$\mathcal{R}^1 = (\mathcal{R}_i^1 + \mathcal{R}^0)/2 \quad (20)$$

$$\text{Level } 1/l = 1 : [F_t^2, \mathcal{R}_t^2] = \text{Trans_encoder}([F_t^1, \mathcal{R}^1]) \quad (21)$$

$$\mathcal{R}^2 = (\mathcal{R}_t^2 + \mathcal{R}^1)/2 \quad (22)$$

$$\text{Level } 2/l = 2 : [F_{t-i}^3, \mathcal{R}_{t-i}^3] = \text{Trans_encoder}([F_{t-i}^2, \mathcal{R}^2]) \quad (23)$$

$$\mathcal{R}^3 = (\mathcal{R}_{t-i}^3 + \mathcal{R}^2)/2 \quad (24)$$

$$\text{Level } 3/l = 3 : [F_{r \rightarrow x}^4, \mathcal{R}_{r \rightarrow x}^4] = \text{Trans_encoder}([F_{r \rightarrow x}^3, \mathcal{R}^3]) \quad (25)$$

$$\mathcal{R}^4 = (\mathcal{R}_{r \rightarrow x}^4 + \mathcal{R}^3)/2 \quad (26)$$

$$\text{Level } 4/l = 4 : [F_{c \rightarrow x}^5, \mathcal{R}_{c \rightarrow x}^5] = \text{Trans_encoder}([F_{c \rightarrow x}^4, \mathcal{R}^4]) \quad (27)$$

$$\mathcal{R}^5 = (\mathcal{R}_{c \rightarrow x}^5 + \mathcal{R}^4)/2 \quad (28)$$

$\text{Trans_encoder}()$ denotes a standard Transformer encoder layer. F_i^l is the image feature embedding in level l . We set $F_i^0 = F_i$, $F_t^0 = F_t$, $F_{t-i}^0 = F_{t-i}$, $F_{r \rightarrow x}^0 = F_{r \rightarrow x}$, $F_{c \rightarrow x}^0 = F_{c \rightarrow x}$.

$$\mathcal{R}^0 = (F_i^0 + F_t^0 + F_{t-i}^0 + F_{r \rightarrow x}^0 + F_{c \rightarrow x}^0)/5 \quad (29)$$

As shown above, the sequence combination $\mathcal{R}$ integrates different features. As shown in Eq. (21), $\mathcal{R}^l$ initially combines the image view, followed by fusing textual information via Transformer encoder layers. To fully integrate information from various embeddings, we iterate the progressive fusion module N times. Finally, after the iterative progressive fusion is completed, we concatenate the sequences in $\mathcal{R}$ to extract fused information through several MLP networks:

$$F_{fused} = \text{MLPs}(\mathcal{R}^N) \quad (30)$$

3.3.1. Theoretical basis of HPT

The principles of Eqs. (19)-(30) can be seen in Fig. 5. Progressive fusion enables gradual feature refinement, avoiding single-step fusion's information dilution. Based on the self-attention mechanism of the Transformer's encoder module, we design Eqs. (19), (21), (23), (25), (27) to simultaneously retain the unique information of each modality/feature while enabling cross-modal fusion. Suppose that $\Psi = \{i, t, t-i, r \rightarrow x, c \rightarrow x\}$, where $\psi \in \Psi$, when $Q = K = V = [F_\psi^l, \mathcal{R}^l]$, the self-attention mechanism preserves the key features within the modality. Meanwhile, the concatenation of F_ψ^l and $\mathcal{R}^l$ ensures the fusion of cross-modal information. In forward propagation, Eqs. (20), 22, 24, 26, 28 can be rewritten in residual form:

$$\mathcal{R}^{l+1} = \mathcal{R}^l + \frac{1}{2}(\mathcal{R}_\psi^{l+1} - \mathcal{R}^l) \quad (31)$$

In back propagation, the gradient becomes:

$$\frac{\partial \mathcal{L}}{\partial \mathcal{R}^l} = \frac{\partial \mathcal{L}}{\partial \mathcal{R}^{l+1}} \frac{\partial \mathcal{R}^{l+1}}{\partial \mathcal{R}^l} = \frac{\partial \mathcal{L}}{\partial \mathcal{R}^{l+1}} (1 + \frac{1}{2}(\frac{\partial(\mathcal{R}_\psi^{l+1} - \mathcal{R}^l)}{\partial \mathcal{R}^l})) \quad (32)$$

where $\mathcal{L}$ denotes the loss function. We can see that the advantages of the aforementioned design are as follows: (1) The average weighting preserves previous fusion results while adds new information; (2) The coefficient 1/2 in Eq. (32) compresses the gradient magnitude of the residual term, preventing explosion; (3) The identity path (gradient = 1) stabilizes backpropagation even if the nonlinear transformation introduces gradient instability.

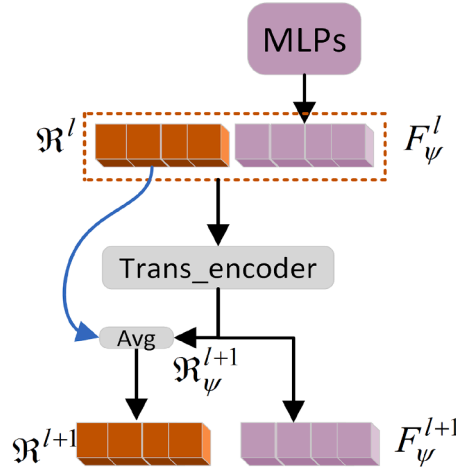


Fig. 5. Graphical representation of Eqs. (20-29).

3.4. Loss function

The fused information F_{fused} is input into the classifier, which consists of a MLP network and a sigmoid output layer. The predicted label $\hat{y}$ can be represented as:

$$\hat{y} = \text{Sigmoid}(\text{MLP}(F_{fused})) \quad (33)$$

Fake news detection is fundamentally a binary decision problem. Thus, the loss function is derived through the following formulation:

$$\mathcal{L}(\theta) = -\frac{1}{D} \sum_{\Gamma} [y \log(p(\hat{y})) + (1 - y) \log(1 - p(\hat{y}))] \quad (34)$$

where θ represents the parameter set in our model. The training set is denoted by Γ containing D samples, with y and $\hat{y}$ representing the true and predicted labels respectively.

3.5. Explanation module

3.5.1. MLIME overview

To explain the influence of each modality and LLM-generated contents on the results of fake news detection, we perform a detailed qualitative analysis for each feature. However, modalities/features can be complementary, and thus it is necessary to consider explainability models that account for the interactions between modalities/features. Therefore, we propose MLIME for the extension of LIME (Ribeiro et al., 2016) in multimodal settings. Moving beyond the black-box nature of complex multimodal models, MLIME provides post-hoc explanations that help users understand why a particular news is classified as real or fake. Currently, mainstream post-hoc methods primarily include SHAP (Lundberg & Lee, 2017) and Grad-CAM (Selvaraju et al., 2017). In the experimental section, we evaluate the effectiveness of SHAP and MLIME. Grad-CAM is designed for visualizing CNN in unimodal (especially image) contexts. Unlike these methods, MLIME is specifically designed for multimodal scenarios. For a given news item, we generate explanations for each feature separately by training explainable models on perturbed samples while keeping the other modalities fixed and incorporating multimodal fusion information.

3.5.2. The process of MLIME

The process for generating an explanation for a news item is as follows: (1) We generate the explanation of text modality on fake news detection. Suppose that G is a set of explainable models, each explanation model $g(\cdot) \in G$ has a measure of complexity, which is denoted as $O(g(\cdot))$. Given a news text T and its representation f_t , we define $h(t)$ as the likelihood that T is assigned to a specified category. π_t is defined as the locality around f_t . The explanation model $g(\cdot)$ for the instance f_t is compared to the original model $h(\cdot)$ by minimizing $\mathcal{L}(h(\cdot), g(\cdot), \pi_t)$, which indicates the degree to which $g(\cdot)$ fails to approximate $h(\cdot)$ accurately in the area surrounding f_t . Thus, the generation of the explanation for instance f_t can be expressed as follows:

$$\xi(f_t) = \operatorname{argmin}_{g \in G} \mathcal{L}(h(\cdot), g(\cdot), \pi_t) + O(g(\cdot)) \quad (35)$$

Eq. (35) defines the optimal explanation model $g(\cdot)$ by minimizing both the approximation error to the original model $h(\cdot)$ and the model complexity. We approximate $\mathcal{L}(h(\cdot), g(\cdot), \pi_t)$ by drawing samples. We sample instances near f_t by randomly selecting nonzero elements of f_t . Given a perturbed instance $z' \in Z$, which includes a portion of nonzero elements of f_t , we retrieve the instance in its original form z , and calculate $h(z)$ with the initial model, which can be considered as the label c for the explanation model $g(\cdot)$. To

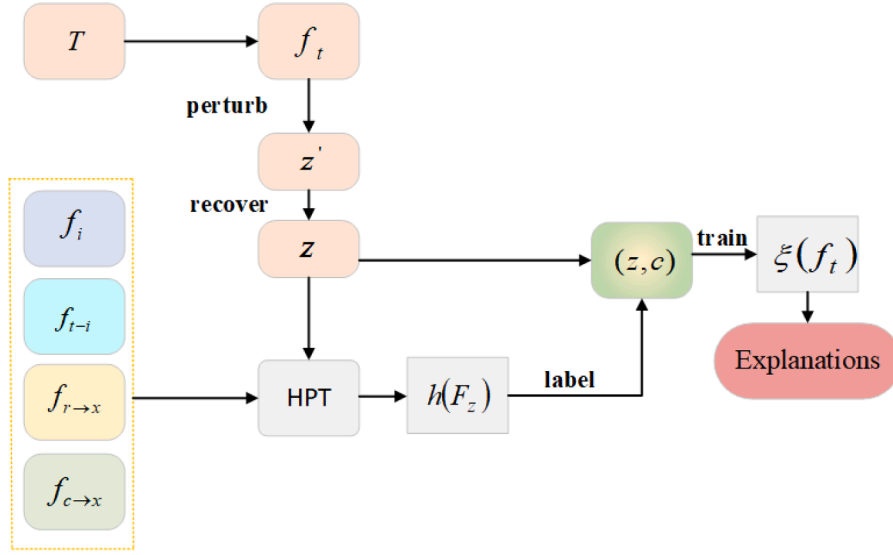


Fig. 6. The flowchart of MLIME for the generation of the explanation on text modality.

derive the explanation for the instance f_t , we optimize Eq. (35) using perturbed samples Z . Before training, the explanation model $g(\cdot)$ is initialized as a Ridge regression model, while $O(g(\cdot))$ is configured as follows:

$$O(g(\cdot)) = \begin{cases} \infty & \text{if } \|w_g\|_0 > \gamma \\ 0 & \text{otherwise} \end{cases} \quad (36)$$

where w_g is the number of non-zero weights in model $g(\cdot)$, γ is a constant value. The theoretical foundation of this formula lies in constraining the complexity of $g(\cdot)$ by enforcing the use of only γ features, which aligns with human cognitive patterns. $\pi_t(z)$ is defined based on cosine distance function:

$$\pi_t(z) = \exp\left(-\frac{(1 - \frac{f_t \cdot z}{\|f_t\| \|z\|})^2}{\sigma^2}\right) \quad (37)$$

Based on the above equation, $\mathcal{L}(h(\cdot), g(\cdot), \pi_t)$ is defined as follows:

$$\mathcal{L}(h(\cdot), g(\cdot), \pi_t) = \sum_{z' \in Z} \pi_t(z) (h(z) - g(z'))^2 \quad (38)$$

(2) Considering that our method involves multiple features of input, we perturb only the text modality, while the other modalities will have the same representations for each z' . Specifically, each perturbed sample z' generates a distinct text representation, while the representations of other features, such as image features, background knowledge, and comments, remain fixed. For each perturbed sample z' and its recovered space z , we fuse all features through the HPT, which can be denoted as $F_z = \text{HPT}(z, f_i, f_{t-i}, f_{r \rightarrow x}, f_{c \rightarrow x})$. Then, we predict each perturbed sample F_z using the original model and obtain the prediction result $c = h(F_z)$. In this way, Eq. (38) should be rewritten as:

$$\mathcal{L}(h(\cdot), g(\cdot), \pi_t) = \sum_{z' \in Z} \pi_t(z) (h(F_z) - g(z'))^2 \quad (39)$$

Eq. (39) is an extended loss function incorporating multimodal fusion (via HPT) to evaluate perturbations while keeping non-text features fixed. In our experiment, we generate 1000 perturbed samples with their corresponding labels for the text instance f_t . The flowchart of MLIME is shown in Fig. 6. It shows the generation of the explanation on text modality.

(3) We perform the same approach on other features to complete the explanation of the influence of each feature on the results of fake news detection. The final output of MLIME is a user-intelligible highlight of the most influential words and image patches. Moreover, we can combine the visualization results from each modality/feature with human cognition to explain the effectiveness of the model's detection results.

4. Experiments

We first describe two datasets and then specify the experimental setup in detail. We then evaluate our method in comparison with state-of-the-art methods. Furthermore, we present ablation analysis, parameter analysis, robustness testing and efficiency evaluations (see Appendix C) to thoroughly discuss the effectiveness of our model. Finally, model explainability is analyzed using MLIME, while the results of feature fusion are visually represented.

Table 2
Detailed information of the news dataset.

Dataset	WeiBo21	FineFake
Language	Chinese	English
Domains	9 (science, military, education, disaster, politics, health, finance, entertainment, and society)	6 (politics, entertainment, business, health, society, and conflict)
Fake News	2987	8595
Real News	3023	6545
Avg. Number of Terms	122.06	1392.07

4.1. Datasets and evaluations

Our model is tested on two public benchmark datasets, which are WeiBo21 (Nan et al., 2021a) and FineFake (Zhou et al., 2024). During data preprocessing, we remove incomplete samples (text-only, image-only, or corrupted image files) that typically introduce noise and hinder cross-modal feature learning. Only news items containing both text and images remain in the dataset to ensure stable cross-modal relationship learning. Additionally, samples with sensitive content that might trigger LLM rejection are excluded to maintain consistent auxiliary information generation. This preprocessing strategy improves data consistency while strengthening the model's generalization and robustness by ensuring complete multimodal representations and reliable auxiliary information generation.

The WeiBo21 dataset is a Chinese dataset, which encompasses news content across nine distinct domains. The original Weibo21 dataset contains 9128 samples, with 6010 valid samples retained after preprocessing:

- (1) Training: 4,808 samples (2,425 real/2,383 fake);
- (2) Testing: 601 samples (294 real/307 fake);
- (3) Validation: 601 samples (304 real/297 fake).

The FineFake dataset is an English dataset, which encompasses news from six domains and eight distinct platforms. Each news item includes various modalities, potential social context, semi-verified common knowledge, and detailed annotations that go beyond simple binary labels. The dataset initially includes 16,909 samples, with 15,140 remaining post-processing. The dataset consists of:

- (1) Training: 9,084 samples (3,858 real / 5226 fake);
- (2) Testing: 3,028 samples (1,331 real / 1697 fake);
- (3) Validation: 3,028 samples (1,356 real / 1672 fake).

Both datasets are summarized in Table 2. We use accuracy (Acc.), precision (Prec.), recall (Rec.), F1-score (F1), and AUC as the evaluation metrics for this task. Besides these two datasets, we also conduct robustness experiments on an LLM-generated fake news dataset. See Appendix D for details.

4.2. Experimental settings

4.2.1. Implementation details

For both Chinese and English text data, the pre-trained BERT models “chinese-bert-wwm-ext” and “bert-base-uncased” are used for embedding representation, with a maximum token length of 170 and an embedding dimension of 768. During training, early stopping is implemented with a patience parameter set to 5. The loss function employed is binary cross-entropy. The classifier is a MLP, with input and hidden layer dimensions of 768 and 384, respectively. We train the model for 50 epochs using a batch size of 128 and optimize the model with Adam. The learning rate is optimized with StepLR (i.e., a learning rate scheduler in PyTorch), where the step_size is set to 100, and gamma is set to 0.98, meaning the learning rate is adjusted every 100 training steps by multiplying the current learning rate by 0.98.

The image feature extraction uses a ViT model, where the input image size is 224×224 , and the output embedding dimension is 768. The image-encoder module in CLIP uses pre-trained ViT architecture, with an input image size of 224×224 and an output embedding dimension of 512. The text-encoder module also produces an embedding dimension of 512.

4.2.2. Baselines

We compare LLM-MFEFND with text-based unimodal methods (TM), multimodal approaches (MM), and LLM-based methods (LLM). We choose DeepSeek and ChatGLM in our experiments for their open-source nature, strengths in Chinese and multilingual processing, and computational efficiency. The features utilized by various methods are presented in Table 3. We also compare DeepSeek and ChatGLM against other open-source LLMs (see Appendix B for details).

TM1: TextCNN (Kim, 2014): The TextCNN model is a highly effective text encoder. The original paper uses word2vec for word embeddings, we utilize BERT embeddings in our experiment.

TM2: BERT (Devlin et al., 2019): The BERT model is a bidirectional encoded representation model based on the transformer architecture. We use BERT to encode the tokens of news content and input the extracted mean embedding vectors into MLP to generate final predictions.

Table 3

Feature composition in baseline experiments.

Category	Method	Text	Image	Domain Information	Aligned Multimodal Feature	Background	Comments
Unimodal Method	TextCNN	✓	×	×	×	×	×
	BERT	✓	×	×	×	×	×
	MoSE	✓	×	×	×	×	×
Multimodal Method	EDDFN	✓	×	✓	×	×	×
	EANN	✓	✓	×	×	×	×
	MDFEND	✓	×	✓	×	×	×
	MMFN	✓	✓	×	×	×	×
	MSACA	✓	✓	×	×	×	×
	MTS	✓	✓	×	×	×	×
LLM-based Method	ARG	✓	×	×	×	✓	×
	LLM-Zero-GLM	✓	×	×	×	×	×
	LLM-Few-GLM	✓	×	×	×	×	×
	LLM-Zero-DS	✓	×	×	×	×	×
	LLM-Few-DS	✓	×	×	×	×	×
Ours	LLM-MFEFND	✓	✓	×	✓	✓	✓

TM3: MoSE (Qin et al., 2020): The MoSE method proposes a novel framework called Mixture of Sequential Experts (MoSE), which effectively handles user text data for fake news detection by combining LSTM networks with the Multi-Gated Mixture of Experts (MMoE) model.

MM1: EDDFN (Silva et al., 2021b): The EDDFN model consists of two key components: unsupervised domain embedding learning and supervised domain-independent classification, jointly extracting domain-specific and cross-domain knowledge.

MM2: EANN (Wang et al., 2018): The EANN model primarily extracts both textual and visual features. An event discriminator is employed to distinguish events and eliminate specific features, while the multimodal feature extractor undergoes adversarial learning with the event discriminator to effectively detect fake news.

MM3: MDFEND (Nan et al., 2021b): The MDFEND model addresses the issue of domain shift (the difference in data distribution across different domains) by integrating multi-domain data. The core of the model lies in using a domain gating mechanism that dynamically selects the most suitable expert networks to extract news representations based on the domain and content features of the news.

MM4: MMFN (Zhou et al., 2023): The MMFN model integrates multi-scale feature extraction, leveraging both fine-grained details and coarse-grained semantics from news content. It processes text and visual inputs using CLIP encoding for global features and a co-attention Transformer for localized patterns.

MM5: MSACA (Wang et al., 2024b): The MSACA model enhances the semantic consistency between text and image data, selecting the most discriminative unimodal and cross-modal embeddings for fake news detection.

MM6: MTS (Sun et al., 2025): The MTS model leverages Taylor series expansion to capture low-order and high-order cross-modal interactions for fake news detection, ensuring linear scalability and enhanced interpretability.

LLM1: ARG (Hu et al., 2024): The ARG model utilizes LLMs to generate multi-perspective, informative rationales to assist smaller language models in fake news detection.

LLM2: LLM-Zero-GLM (GLM et al., 2024): The LLM-Zero-GLM model employs Zero-Shot Prompting based on ChatGLM (GLM-4-Air), constructing prompts that consist solely of the task description and the provided news.

LLM3: LLM-Few-GLM (GLM et al., 2024): The LLM-Few-GLM model utilizes Few-Shot Prompting based on ChatGLM (GLM-4-Air), which gives the model a few labeled examples to guide its understanding of the task and improve its performance. We adopt a 4-shot prompting strategy consisting of two authentic samples and two fake samples.

LLM4: LLM-Zero-DS (Liu et al., 2024): The LLM-Zero-DS model employs Zero-Shot Prompting based on DeepSeek (DeepSeek-v3).²

LLM5: LLM-Few-DS (Liu et al., 2024): The LLM-Few-DS model utilizes Few-Shot Prompting based on DeepSeek (DeepSeek-v3).

OUR1 : LLM-MFEFND-GLM: The LLM-MFEFND model utilizes ChatGLM (GLM-4-Air) for augmented feature generation.

OUR2 : LLM-MFEFND-DS: The LLM-MFEFND model utilizes DeepSeek (DeepSeek-v3) for augmented feature generation.

The parameters for the baseline models are configured as follows:

TextCNN, BERT, MoSE: The dimensions of the MLP (classifier) are $\{\{320,384,0.2\},\{768,384,0.2\},\{320,384,0.2\}\}$, where 0.2 is the parameter of the dropout layer. batch size = 64, epochs = 50, early_stop = {NaN, 3, 5};

EDDFN, EANN, MDFEND, MMFN, MSACA: The dimensions of the MLP (classifier) are $\{\{320,384,0.2\},\{320,384,0.2\},\{320,384,0.2\},\{48,48,2\},\{256,64,2\}\}$, batch size = 64, epochs = {50,50,50,50,100}, early_stop = {5,5,5,5,NaN};

TextCNN: The output channels stay constant at 64 across all kernel sizes: {1, 2, 3, 5, 10};

² <https://github.com/deepseek-ai>

Table 4

Comparison of LLM-MFEFND with the latest approaches on WeiBo21. The best results are bolded. (*: open-source). + ($p \leq 0.05$), ++ ($p \leq 0.005$), indicating statistical significance via paired t -test compared with the best baseline.

Category	Method	AUC	Acc.	Fake News			Real News		
				Prec.	Rec.	F1	Prec.	Rec.	F1
Unimodal Method	TextCNN	0.944	0.845	0.796	0.909	0.833	0.922	0.798	0.856
	BERT	0.935	0.866	0.840	0.848	0.844	0.887	0.880	0.883
	MoSE*	0.946	0.901	0.904	0.858	0.880	0.899	0.932	0.915
Multimodal Method	EDDFN*	0.931	0.871	0.834	0.868	0.851	0.900	0.873	0.886
	EANN*	0.952	0.884	0.882	0.838	0.860	0.884	0.917	0.901
	MDFEND*	0.980	0.928	0.900	0.959	0.929	0.958	0.899	0.927
	MMFN*	0.977	0.925	0.938	0.911	0.924	0.911	0.939	0.925
	MSACA*	0.952	0.905	0.910	0.894	0.902	0.898	0.914	0.906
	MTS*	0.979	0.928	0.898	0.963	0.929	0.962	0.896	0.927
LLM-based Method	ARG*	0.965	0.911	0.907	0.916	0.912	0.915	0.905	0.910
	LLM-Zero-GLM	0.621	0.613	0.802	0.329	0.467	0.563	0.914	0.696
	LLM-Few-GLM	0.659	0.652	0.589	0.934	0.723	0.861	0.384	0.532
	LLM-Zero-DS	0.812	0.811	0.779	0.861	0.817	0.850	0.763	0.804
	LLM-Few-DS	0.716	0.711	0.632	0.976	0.767	0.952	0.456	0.617
Ours	LLM-MFEFND-GLM	0.982	0.943 ⁺	0.930	0.955	0.942 ⁺⁺	0.956	0.931	0.943 ⁺
	LLM-MFEFND-DS	0.983	0.945 ⁺	0.945	0.942	0.944 ⁺⁺	0.945	0.948 ⁺⁺	0.946 ⁺

EANN: The output channels stay constant at 64 across all kernel sizes: {1, 2, 3, 5, 10}, and the parameter α is defined as follows:

$$\alpha = \max \left(\frac{2}{1 + \exp \left(-\frac{10 \cdot \text{epoch}}{\text{epochs}} \right)} - 1, 0.1 \right) \quad (40)$$

where epoch represents the current training round, epochs represents the total number of training rounds.

ARG: The dimension of the classifier is {320,348,0.2}, the dimension of the co-attention network is 300, batch size = 64, epochs = 50.

MTS: The training uses a batch size of 24 and limits input sequences to 200 tokens. The learning rate is fixed at 2e-5 to ensure stable training.

4.3. Overall performance

To evaluate our proposed method from an overall perspective, we compare LLM-MFEFND with various state-of-the-art unimodal and multimodal fake news detection methods on two datasets, respectively. We also compare LLM-MFEFND with LLM-based methods to evaluate the effectiveness of integrating LLMs into the fake news detection task. Tables 4 and 5 present a comparison between LLM-MFEFND and other methods. Figs. 7 and 8 display heatmaps of the average performance for all methods.

The data presented in Tables 4 and 5, as well as Figs. 7 and 8, demonstrate that LLM-MFEFND outperforms all other models in terms of overall accuracy, confirming the effectiveness of our proposed method. Notably, LLM-MFEFND-DS achieves 94.5% accuracy and 0.945 F1-score on WeiBo21, along with 81.1% accuracy and 0.807 F1-score on FineFake. Additionally, these results confirm that the choice of LLMs affects model performance. While both models show comparable results on the Chinese dataset, DeepSeek shows superior performance over ChatGLM on the English dataset.

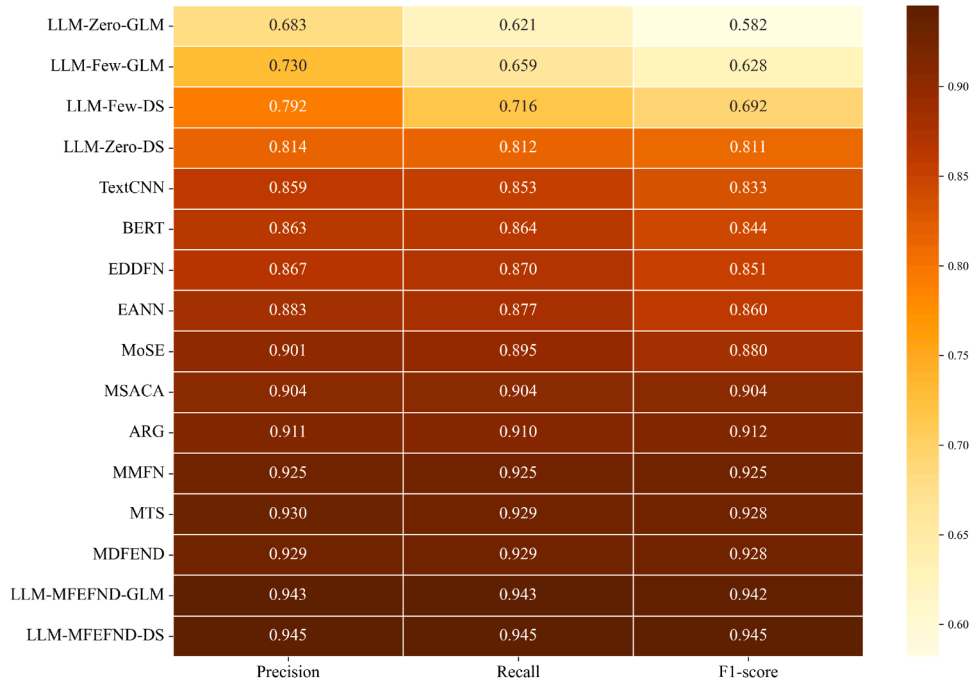
In unimodal methods, MoSE performs better than TextCNN and BERT, this is because MoSE uses a shared expert pool with multiple independent principals targeting different domains. TextCNN and BERT only extract features from the text and cannot understand the information in the image. Many instances of fake news are misleading due to the images they use, and models that rely solely on textual information often struggle to capture these multimodal cues.

In multimodal methods, MTS and MMFN outperform MDFEND, MSACA, EANN, and EDDFN. The superior performance of MTS and MMFN compared to other methods can be attributed to their advanced multimodal representation learning mechanisms. Specifically, MMFN enhances feature extraction by processing multimodal inputs at varying granularities. MTS can generate deeper interactions between texts and images by multimodal Taylor series. MDFEND leverages a soft-sharing mechanism to aggregate cross-domain knowledge, yielding better performance compared to EANN and EDDFN. However, MDFEND primarily focuses on cross-domain knowledge and struggles to effectively fuse cross-modal information, which limits its performance. In some cases, the semantics of text and images may not fully align, leading to suboptimal model performance. Although MSACA reduces semantic ambiguity between text and images through VAE model, cross-modal semantic alignment remains a challenging problem. Furthermore, MSACA shows a notable disparity with a high fake news F1 (0.808) but a significantly lower real news F1 (0.742). In contrast, LLM-MFEFND-DS performs well on both classes. In Figs. 7 and 8, when considering macro-averaged metrics that aggregate performance across both classes equally, our method demonstrates superior overall performance. In LLM-based methods, ARG performs better than

Table 5

Comparison of LLM-MFEFND with the latest approaches on FineFake. The best results are bolded. (*: open-source). + ($p \leq 0.05$), ++ ($p \leq 0.005$), indicating statistical significance via paired *t*-test compared with the best baseline.

Category	Method	AUC	Acc.	Fake News			Real News		
				Prec.	Rec.	F1	Prec.	Rec.	F1
Unimodal Method	TextCNN	0.830	0.753	0.750	0.656	0.700	0.754	0.829	0.790
	BERT	0.848	0.764	0.727	0.742	0.735	0.794	0.782	0.788
	MoSE*	0.862	0.776	0.760	0.718	0.738	0.788	0.822	0.804
Multimodal Method	EDDFN*	0.824	0.746	0.766	0.607	0.677	0.735	0.854	0.790
	EANN*	0.858	0.773	0.773	0.686	0.727	0.773	0.842	0.806
	MDFEND*	0.858	0.776	0.806	0.646	0.717	0.760	0.878	0.814
	MMFN*	0.866	0.789	0.780	0.762	0.752	0.796	0.839	0.817
	MSACA*	0.834	0.781	0.777	0.842	0.808	0.785	0.705	0.742
	MTS*	0.875	0.788	0.759	0.758	0.758	0.810	0.811	0.811
LLM-based Method	ARG*	0.866	0.776	0.795	0.806	0.800	0.752	0.738	0.745
	LLM-Zero-GLM	0.662	0.640	0.794	0.486	0.603	0.559	0.838	0.671
	LLM-Few-GLM	0.685	0.672	0.779	0.581	0.666	0.595	0.789	0.678
	LLM-Zero-DS	0.692	0.672	0.811	0.561	0.663	0.581	0.823	0.681
	LLM-Few-DS	0.672	0.669	0.742	0.651	0.694	0.595	0.693	0.640
Ours	LLM-MFEFND-GLM	0.881 ++	0.800 ++	0.797	0.730	0.762	0.801	0.854	0.827 +
	LLM-MFEFND-DS	0.882 ++	0.811 ++	0.811 ++	0.745	0.776	0.812 +	0.864	0.837 +

**Fig. 7.** Performance heatmap visualization for WeiBo21.

other LLM-based methods. This may be because ARG uses LLMs to provide multi-perspective reasoning, enhancing the recognition of smaller models. The performance of DeepSeek is superior to that of ChatGLM, which may be due to DeepSeek's use of broader knowledge coverage and a richer, more diverse dataset of fake news, encompassing more linguistic styles. Both LLM-Zero and LLM-Few directly use LLMs to determine the authenticity of news, but fail to achieve effective results. This result also indicates that LLMs cannot directly replace models for fake news detection. Although few-shot settings are generally believed to improve performance by providing task examples, we observe that DeepSeek performs better in zero-shot than in few-shot. This may be due to DeepSeek's stronger zero-shot reasoning ability. During pretraining, it integrates high-quality instruction fine-tuning data and generalization training, enabling accurate judgment based on input context and internal knowledge without explicit examples. In zero-shot, the model can efficiently reason based on task descriptions, while in few-shot, added examples interfere with its reasoning path, leading to decreased performance. This suggests that when using LLMs, one should choose the zero-shot or few-shot strategy flexibly based on the model's characteristics.

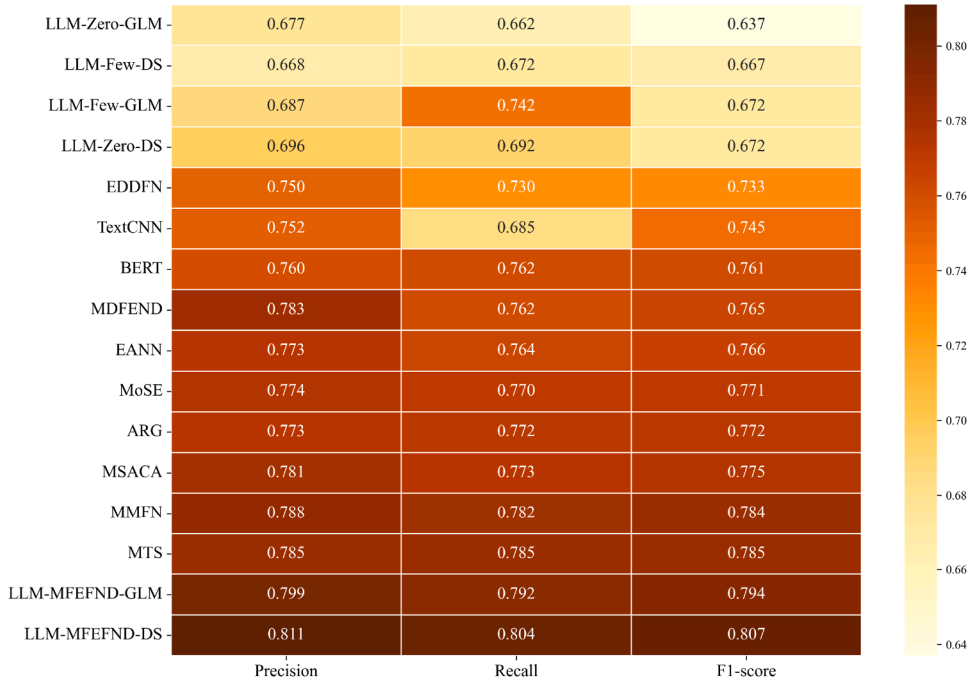


Fig. 8. Performance heatmap visualization for FineFake.

LLM-MFEFND has three major advantages over the above models: (1) LLM-MFEFND extracts textual and visual features from multi-views to ensure the comprehensiveness of news features; (2) LLM-MFEFND uses LLM to generate background and comments, greatly enriching the semantic information; (3) LLM-MFEFND realizes efficient progressively cross-feature fusion through the HPT and improves the overall performance.

4.4. Ablation studies

4.4.1. Ablation study on feature components

To evaluate individual component contributions in LLM-MFEFND, we perform ablation studies through sequential removal of each component. The experimental setup is outlined below:

- (1) W/o T: Remove the component of text feature extraction; with the textual modality removed, the news-background-comments interaction module is effectively disabled. Consequently, the background and comment features are processed solely through attention mechanisms before being directly propagated to the HPT.
- (2) W/o V: Remove the component of image feature extraction; the other four features are fed into the HPT module.
- (3) W/o B: Remove the background knowledge; the textual features no longer interact with background, while the other four features are fed into the HPT module.
- (4) W/o F: Remove the aligned multimodal feature; the other four features are fed into the HPT module.
- (5) W/o C: Remove the comments; the textual features no longer interact with comments, while the other four features are fed into the HPT module.
- (6) W/o U: Remove the news-background-comments interaction module; instead, the background and comment features (extracted by BERT) are pooled and fed directly into HPT.

As illustrated in [Tables 6 and 7](#), the results of each ablation experiment are presented. It can be observed from the experimental results that the original LLM-MFEFND-GLM and LLM-MFEFND-DS outperform all the ablation variants, with the introduction of each module significantly improving the model performance. This proves the effectiveness of each module in the model. The key findings are as follows:

- (1) Excluding the text encoder results in the most significant drop in accuracy and AUC, with a 3.0% and 0.5% decrease on WeiBo21 using GLM, and 3.3% and 2.6% with DS, respectively. This highlights the critical role of text features in fake news detection.
- (2) Removing the module of image feature extraction results in a noticeable accuracy drop on two datasets using GLM and DS. This highlights the importance of image features in detection.
- (3) Omitting the aligned multimodal module results in accuracy drops of 2.8% on WeiBo21 and 1.3% on FineFake using GLM, and 1.5% and 3.7% with DS. This indicates that extracting features from both text and images enhances detection performance.

Table 6
Ablation study on two datasets using ChatGLM.

Dataset	Method	AUC	Acc.	Fake News			Real News		
				Prec.	Rec.	F1	Prec.	Rec.	F1
WeiBo21	LLM-MFEFND-GLM	0.982	0.943	0.930	0.955	0.942	0.956	0.931	0.943
	w/o T	0.977	0.913	0.926	0.901	0.913	0.902	0.926	0.914
	w/o V	0.980	0.921	0.882	0.969	0.923	0.967	0.923	0.919
	w/o F	0.976	0.915	0.888	0.945	0.915	0.944	0.885	0.914
	w/o B	0.979	0.936	0.902	0.976	0.937	0.975	0.899	0.935
	w/o C	0.978	0.933	0.917	0.948	0.933	0.949	0.918	0.933
	w/o U	0.971	0.925	0.897	0.955	0.925	0.954	0.895	0.924
FineFake	LLM-MFEFND-GLM	0.881	0.800	0.797	0.730	0.762	0.801	0.854	0.827
	w/o T	0.873	0.791	0.800	0.701	0.747	0.786	0.862	0.822
	w/o V	0.880	0.791	0.729	0.832	0.777	0.852	0.758	0.802
	w/o F	0.868	0.787	0.766	0.741	0.753	0.802	0.822	0.812
	w/o B	0.863	0.779	0.766	0.718	0.741	0.789	0.828	0.808
	w/o C	0.879	0.788	0.719	0.849	0.778	0.862	0.739	0.796
	w/o U	0.860	0.779	0.743	0.762	0.752	0.809	0.793	0.801

Table 7
Ablation study on two datasets using DeepSeek.

Dataset	Method	AUC	Acc.	Fake News			Real News		
				Prec.	Rec.	F1	Prec.	Rec.	F1
WeiBo21	LLM-MFEFND-DS	0.983	0.945	0.945	0.942	0.944	0.945	0.948	0.946
	w/o T	0.957	0.912	0.893	0.932	0.912	0.932	0.893	0.912
	w/o V	0.980	0.938	0.924	0.952	0.938	0.953	0.925	0.939
	w/o F	0.974	0.930	0.938	0.918	0.928	0.923	0.941	0.932
	w/o B	0.979	0.933	0.918	0.949	0.933	0.949	0.919	0.934
	w/o C	0.973	0.931	0.904	0.963	0.932	0.962	0.902	0.931
	w/o U	0.979	0.928	0.934	0.924	0.929	0.923	0.933	0.928
FineFake	LLM-MFEFND-DS	0.882	0.811	0.811	0.745	0.776	0.812	0.864	0.837
	w/o T	0.873	0.789	0.781	0.723	0.751	0.794	0.841	0.817
	w/o V	0.877	0.792	0.763	0.763	0.763	0.814	0.814	0.814
	w/o F	0.862	0.774	0.741	0.748	0.744	0.801	0.795	0.798
	w/o B	0.876	0.792	0.759	0.772	0.765	0.819	0.808	0.813
	w/o C	0.875	0.788	0.759	0.758	0.759	0.811	0.795	0.811
	w/o U	0.825	0.789	0.840	0.653	0.735	0.762	0.899	0.825

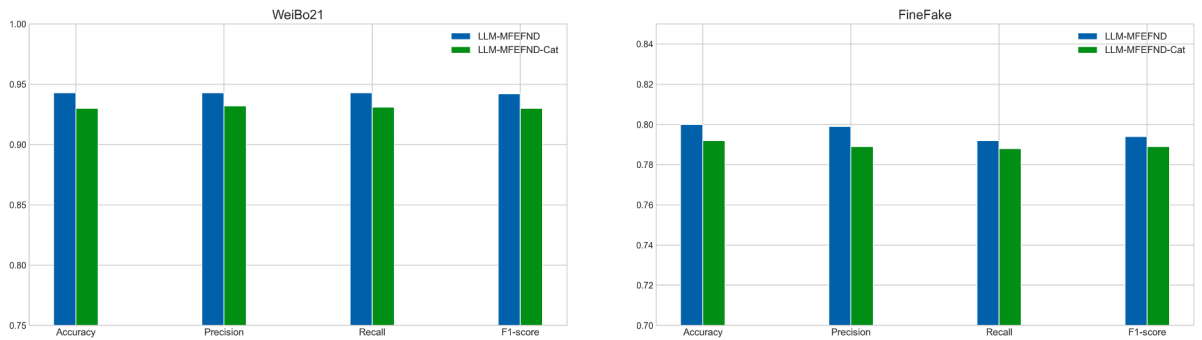


Fig. 9. Comparative study of different fusion methods.

- (4) The elimination of the background module causes accuracy reductions of 0.7% and 2.1% on WeiBo21 and FineFake using GLM, and 1.2% and 1.9% with DS. This demonstrates the importance of the background module in improving model precision.
- (5) Removing the comments module leads to a 1.0% and 1.2% decrease in accuracy on the two datasets using GLM, and 1.4% and 2.3% with DS, highlighting its substantial impact on detection performance.
- (6) Excluding the interaction module results in accuracy drops of 1.8% and 2.1% using GLM, and 1.7% and 2.2% with DS. This suggests that combining news, background, and comments features provides additional semantic information, boosting model performance.

Table 8
Experimental results of different fusion orders.

Dataset	Fusion Order	AUC	Acc.	F1
WeiBo21	$F_l, F_i, F_{l-i}, F_{r \rightarrow X}, F_{c \rightarrow X}$	0.983	0.945	0.945
	$F_l, F_i, F_{l-i}, F_{r \rightarrow X}, F_{c \rightarrow X}$	0.973	0.940	0.940
	$F_{r \rightarrow X}, F_l, F_i, F_{l-i}, F_{c \rightarrow X}$	0.983	0.938	0.938
	$F_{c \rightarrow X}, F_{r \rightarrow X}, F_l, F_i, F_{l-i}$	0.979	0.935	0.935
FineFake	$F_l, F_i, F_{l-i}, F_{r \rightarrow X}, F_{c \rightarrow X}$	0.882	0.811	0.807
	$F_l, F_i, F_{l-i}, F_{r \rightarrow X}, F_{c \rightarrow X}$	0.892	0.804	0.804
	$F_{r \rightarrow X}, F_l, F_i, F_{l-i}, F_{c \rightarrow X}$	0.882	0.796	0.795
	$F_{c \rightarrow X}, F_{r \rightarrow X}, F_l, F_i, F_{l-i}$	0.875	0.798	0.796

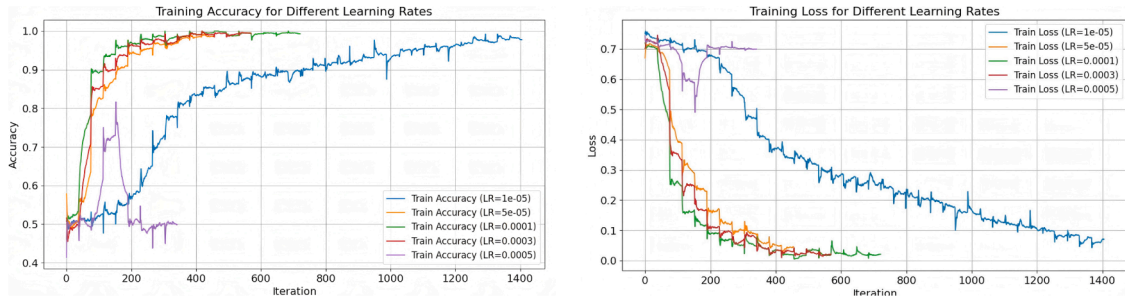


Fig. 10. The training accuracy and loss value under different learning rates on WeiBo21.

4.4.2. Ablation study on fusion module

To evaluate the performance of the HPT module, we compare it with the concatenation method. As shown in Fig. 9, on the WeiBo21 dataset, the HPT method outperforms concatenation method by 1.3% and 1.1% in accuracy and precision, respectively. Comparative evaluation on the FineFake dataset reveals that the HPT technique yields improvements of 0.8% (accuracy) and 1% (precision) compared to concatenation. We believe that HPT not only captures long-range dependencies within the text modality, but also progressively captures complex interactions between modalities. This is particularly advantageous in complex tasks with more than two features. In contrast, the traditional concatenation method is often more efficient. However, when dealing with greater number of features, this approach fails to directly capture the complex relationships between different features.

4.4.3. Ablation study on fusion order

As shown in Table 8, despite the existence of a large number of possible orders, we select several critical alternative schemes for comparison, such as context-first and core-feature-first fusion. The results indicate that while different fusion orders have a certain impact on model performance, the effect is not substantial. Even with varying orders, the performance remains superior to other baselines. On both datasets, the optimal fusion order is consistently $F_l, F_i, F_{l-i}, F_{r \rightarrow X}, F_{c \rightarrow X}$. Notably, fusing images before text reduces performance. This may suggest that text features are more important than visual features, consistent with our ablation study (Section 4.4.1).

Overall, these results confirm that the most effective strategy is to first fuse the core features, namely text and image with their aligned features, and then gradually integrate contextual information generated by LLMs.

4.5. Parameter analysis

4.5.1. Learning rate

In this section, we perform LLM-MFEFND training with varying learning rates. Fig. 10 shows the learning curves of accuracy and loss under different learning rates on the WeiBo21 dataset. The model exhibits stable convergence across different learning rates, with optimal performance observed at 0.0001, yielding the highest accuracy and training efficiency. While convergence still occurs at 1e-5, it demands substantially more iterations. However, a learning rate of 0.0003 proves less effective for this task.

Fig. 11 depicts the learning curves of accuracy and loss under different learning rates on the FineFake dataset. Both the model's training efficiency and accuracy are maximized with the learning rate of 0.0002.

4.5.2. Iteration round N in HPT

We evaluate the effect of interaction rounds (N) in the HPT module by conducting experiments using DeepSeek and ChatGLM on two datasets. We conduct an experiment with varying numbers of interaction rounds (ranging from 1 to 8). As shown in Fig. 12, the results show that in both language scenarios, the models achieve optimal F1-scores when $N = 4$, indicating that an appropriate number of multi-round interactions enhances the models' ability to integrate diverse features, thus improving fake news detection performance. However, when $N > 4$, performance slightly degrades, suggesting that excessive interaction rounds may introduce redundant or noisy information, which could weaken the fusion effectiveness of the HPT.

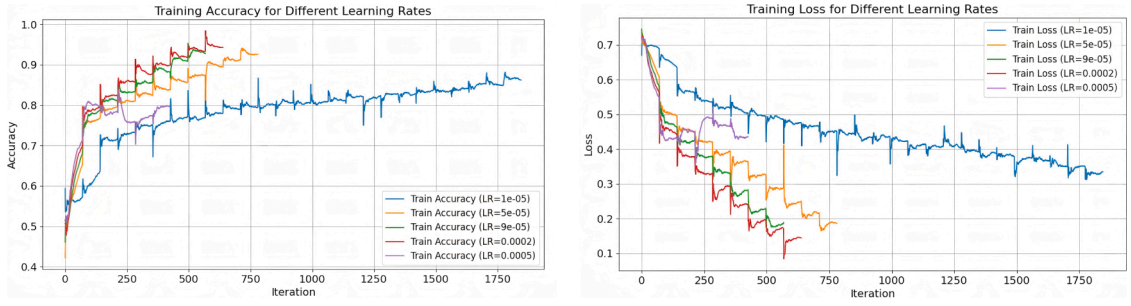


Fig. 11. The training accuracy and loss value under different learning rates on FineFake.

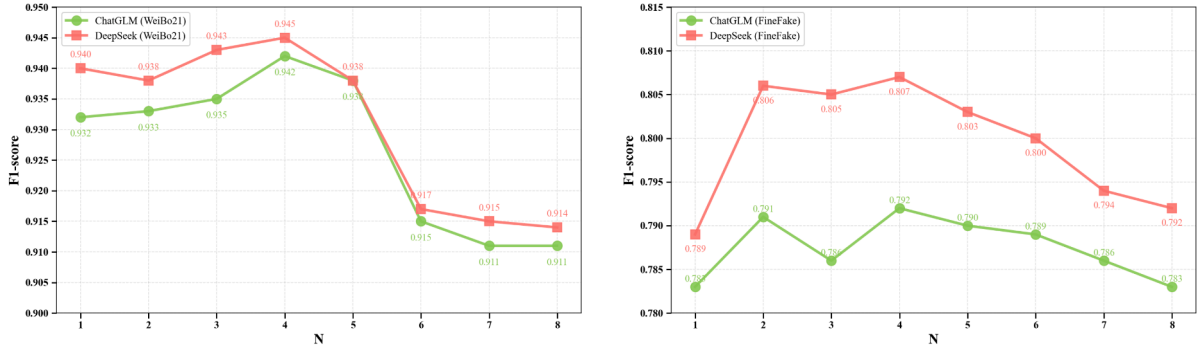


Fig. 12. Impact of iteration round (N) on fake news detection performance.



Fig. 13. Performance of different methods under different fake/real news ratios on WeiBo21.

4.6. Robustness analysis

As shown in Fig. 13 and Fig. 14, we conduct experiments on the FineFake and Weibo21 datasets with varying ratios of fake to real news (e.g., 1:1, 2:1, 1.5:1, 1:2, etc.), evaluating its robustness and generalization capabilities.

On the FineFake dataset, the model demonstrates good performance across various ratios. The F1 consistently ranges between 0.802 and 0.808, with a very small fluctuation range (± 0.006), indicating stable classification performance and insensitivity to changes in class ratios. For example, the F1 is 0.804 at 1:1 ratio and slightly increases to 0.808 at 2:1 ratio. Meanwhile, the AUC remains stable between 0.877 and 0.886. It confirms that the model effectively captures discriminative features between real and fake news even under imbalanced class distributions, without overfitting the majority class.

On the WeiBo21 dataset, the model achieves peak performance at 1:1 ratio (F1: 0.945, AUC: 0.983). When the class ratio adjusts to 2:1 or 1:1.5, performance slightly declines but remains strong (F1 > 0.928, AUC > 0.973). This minor decline is primarily due to



Fig. 14. Performance of different methods under different fake/real news ratios on FineFake.

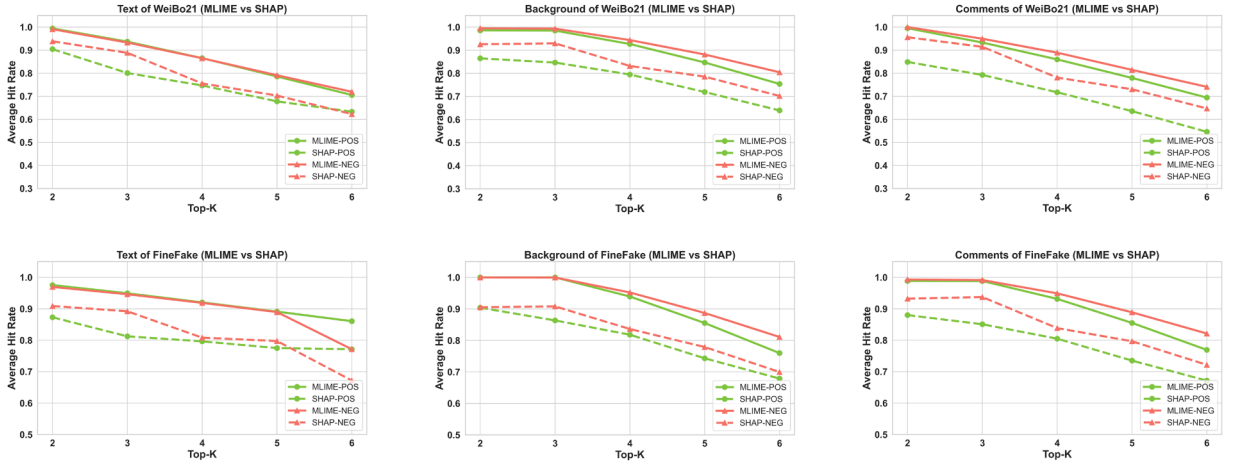


Fig. 15. Comparative quantitative analysis of MLIME and SHAP..

the smaller scale of the WeiBo21 dataset, where class imbalance amplifies sample disparities and causes slight decision boundary shifts. Nonetheless, the model maintains robust performance across all ratios, demonstrating consistent effectiveness on Chinese data.

Additionally, we select MMFN and MTS, which perform well on FineFake, as well as MDFEND and MTS, which perform well on WeiBo21, for comparison with our method. On the FineFake dataset, when the class distribution is imbalanced (e.g., 2:1 or 1:2), both MMFN and MTS exhibit a significant decline in F1 and Accuracy, fluctuating between 0.762 and 0.788. In contrast, our method consistently maintains F1 and Accuracy above 0.800 across varying ratios, demonstrating smaller performance fluctuations and stronger stability. Similarly, our method outperforms MDFEND and MTS with superior results and smaller variations on the WeiBo21 dataset. Overall, compared to other methods, our approach exhibits better robustness and generalization capability.

4.7. Explainability analysis

4.7.1. Quantitative analysis

To quantify the explainability of MLIME, we randomly select a set of 400 correctly detected news items from the test set. Since image interpretations are difficult to annotate, particularly for small patches, we focus on textual information (e.g., text, background, comments) and have it manually labeled by news professionals. Based on MLIME's explanation results, we calculate the explanation hit rate as follows:

$$\text{Average_Hit_Rate}(K) = \frac{1}{M} \sum_{i=1}^M \frac{|\text{TopK_Words}_i \cap \text{GroundTruth_Words}_i|}{K} \quad (41)$$

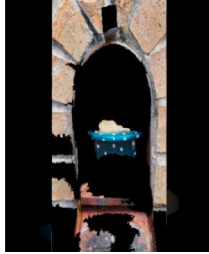
Image	  
	Original image Explanation by SHAP Explanation by MLIME
Text	MLIME: A woman living in Houston, Texas, successfully baked bread in her mailbox during a heat wave. SHAP: A woman living in Houston, Texas, successfully baked bread in her mailbox during a heat wave.
Back-ground	MLIME: Houston, Texas, is known for its hot and humid climate, especially during the summer months. The average high temperature in the summer can exceed 90°F (32°C). Mailboxes in the United States are typically made of metal or other materials that can conduct heat, and they are not designed to maintain temperatures suitable for baking. Bread baking typically requires a consistent internal temperature of around 350-375°F (175-190°C) for a duration of time, which is not achievable in a mailbox exposed to the elements. Furthermore, the insulation and heat retention of a mailbox are not sufficient for baking bread, and the uneven heat distribution would likely result in an improperly cooked product. SHAP: Houston, Texas, is known for its hot and humid climate, especially during the summer months. The average high temperature in the summer can exceed 90°F (32°C). Mailboxes in the United States are typically made of metal or other materials that can conduct heat, and they are not designed to maintain temperatures suitable for baking. Bread baking typically requires a consistent internal temperature of around 350-375°F (175-190°C) for a duration of time, which is not achievable in a mailbox exposed to the elements. Furthermore, the insulation and heat retention of a mailbox are not sufficient for baking bread, and the uneven heat distribution would likely result in an improperly cooked product.
Comments	MLIME: This is such an interesting story! It's a brilliant example of creativity and survival instinct during a heatwave. While it may seem a bit crazy, it shows how people can adapt and make use of what's around them. Who knew a mailbox could be used for baking bread! This sounds ridiculous! Sure, heatwaves can be extreme, but using a mailbox to bake bread is just impractical. It's not safe at all and could easily cause a fire or disrupt the mail system. Definitely not something to encourage." SHAP: This is such an interesting story! It's a brilliant example of creativity and survival instinct during a heatwave. While it may seem a bit crazy, it shows how people can adapt and make use of what's around them. Who knew a mailbox could be used for baking bread! This sounds ridiculous! Sure, heatwaves can be extreme, but using a mailbox to bake bread is just impractical. It's not safe at all and could easily cause a fire or disrupt the mail system. Definitely not something to encourage."

Fig. 16. Results of MLIME and SHAP on one fake news from FineFake dataset.

where $GroundTruth_Words$ represents the set of key feature words corresponding to the annotated category (either fake or real) in human labeling. $M = 400$, $TopK_Words$ represents the set of top-K words in MLIME's explanation output. $|TopK_Words_i \cap GroundTruth_Words_i|$ represents the number of matching words between MLIME's top-ranked explanatory outputs and the ground truth. As shown in Fig. 15, when K is relatively small, the model's explanation hit rate remains above 93%. As K increases, the average hit rate demonstrates a decreasing trend. This is primarily because higher-ranked explanatory terms more accurately reflect the model's core features of attention. Even when $K = 6$, the explanation hit rate remains about 70% ~ 86%.

Furthermore, we implement a SHAP-based explanation method designed for multimodal fake news inputs. Given the absence of a native multimodal version of SHAP, our implementation processes each modality independently. The results show that MLIME consistently outperforms SHAP in explanation hit rates on two datasets. MLIME better captures the interaction characteristics of multi-source text, whereas SHAP relies on feature independence assumptions and additive attribution through sampling approximations. Consequently, MLIME produces more stable explanations and demonstrates superior explanatory reliability.

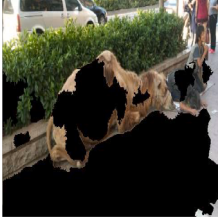
Image	  
Text	Original image Explanation by SHAP Explanation by MLIME MLIME: 【Online Reports of Camels Begging in Multiple Places: Camels' Limbs Were Amputated】 Recently, a netizen @XiamenFoot reported that beggars were seen leading camels to beg on the streets of Foshan, Guangdong. The camels had their limbs amputated at the knees and could only lie on the ground, which is very pitiful. Since the beginning of this year, instances of camels being used for begging have appeared in various places across our country. We call on the relevant authorities to rescue the camels that are being forced to beg in different locations! Please spread the word! Online reports of camels begging in multiple places: Camels' limbs were amputated (photo collection). SHAP: 【Online Reports of Camels Begging in Multiple Places: Camels' Limbs Were Amputated】 Recently, a netizen @XiamenFoot reported that beggars were seen leading camels to beg on the streets of Foshan, Guangdong. The camels had their limbs amputated at the knees and could only lie on the ground, which is very pitiful. Since the beginning of this year, instances of camels being used for begging have appeared in various places across our country. We call on the relevant authorities to rescue the camels that are being forced to beg in different locations! Please spread the word! Online reports of camels begging in multiple places: Camels' limbs were amputated (photo collection).
Back-ground	MLIME: In recent years, there have indeed been reports of animal begging phenomena in some regions, including the use of camels for begging. However, news regarding "camels' limbs being amputated" should be approached with caution, as it may involve animal abuse and illegal activities. Typically, such reports quickly attract public attention and prompt relevant law enforcement agencies to intervene and investigate. In China, animal protection laws are gradually being improved, and there are clear penalties for acts of animal cruelty. SHAP: In recent years, there have indeed been reports of animal begging phenomena in some regions, including the use of camels for begging. However, news regarding "camels' limbs being amputated" should be approached with caution, as it may involve animal abuse and illegal activities. Typically, such reports quickly attract public attention and prompt relevant law enforcement agencies to intervene and investigate. In China, animal protection laws are gradually being improved, and there are clear penalties for acts of animal cruelty.
Comments	MLIME: This is so cruel, I hope the authorities will investigate and deal with it as soon as possible. I've seen similar things on the news before and it's really sad. I hope this is fake news, such behaviour is really intolerable. If it is true, these people should be punished severely by the law. We should raise our awareness of animal protection and not let this happen again. SHAP: This is so cruel, I hope the authorities will investigate and deal with it as soon as possible. I've seen similar things on the news before and it's really sad. I hope this is fake news, such behaviour is really intolerable. If it is true, these people should be punished severely by the law. We should raise our awareness of animal protection and not let this happen again.

Fig. 17. Results of MLIME and SHAP on one fake news from WeiBo21 dataset.

4.7.2. Case study

Figs. 16 and 17 provide two examples of MLIME and SHAP on WeiBo21 and FineFake, respectively. We can observe that the LLM-generated background and comments provide not only supplementary factual context but also multi-perspective insights, thereby providing the model with rich contextual information and extra clues.

Words highlighted in red represent positive contributions to the detection of fake news, while those in blue indicate negative contributions. MLIME explanation results show that in an example from WeiBo21, words that clearly convey violence and exaggeration

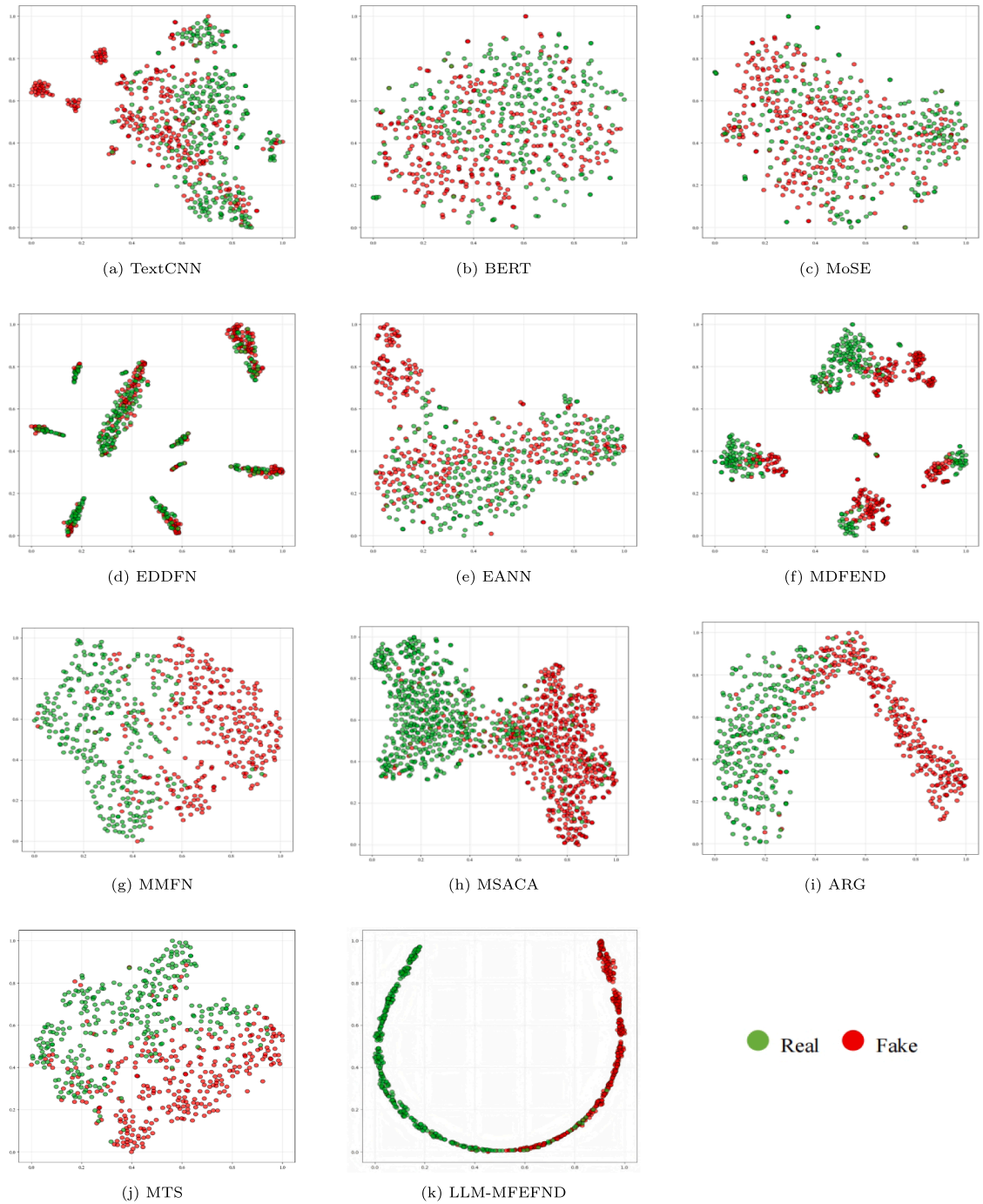


Fig. 18. Feature visualization of each representation method on WeiBo21.

(e.g., “amputated”), as well as negative and doubtful sentiments (e.g., “cruel,” “intolerable”), positively influence the model’s determination of the news as fake. For image features, we see that the characteristics of both the camel and the pedestrian are retained, and these features positively contribute to the assessment of the news’s falsity. Similarly, in FineFake, the positive contributions from the words indeed encompass a rich semantic field related to fake news, such as “baked,” “bread,” and “ridiculous”. This indicates that the words contributing positively to the judgment of fake news contain key semantics related to false information. The example retains the image region of “toasting bread with an Email inbox,” demonstrating that the model effectively highlights this visual element while maintaining consistency with the absurd semantics in the text.

MLIME significantly outperforms SHAP in locating important cues and establishing cross-modal associations. SHAP's visual explanations often neglect critical context in favor of irrelevant backgrounds, while its textual attributions diverge from human cognition and lack cross-modal correlation. For instance, on the FineFake dataset, SHAP misses key positive words like "baked" and "bread," and fails to clearly resolve objects like "mailbox" in the image. Similarly, on WeiBo21, it overlooks critical terms like "beggars" that correspond to visual content. In contrast, MLIME provides more focused visual explanations by targeting event-related subjects. Its textual attributions align closely with human intuition, achieving robust semantic alignment with the image.

4.8. Visualization analysis

Feature visualization is an interpretation method for the fused features. Using t-SNE, we project the fused features into a two-dimensional space for visualization.

As shown in Fig. 18 and 19, the final visualization results indicate that, compared to other methods, our HPT approach better clusters news of the same type together. Moreover, the clusters generated by our method show sharper boundaries and less overlap. Specifically, on the WeiBo21 dataset, LLM-MFEFND's feature distribution forms a distinct crescent-shaped structure, achieving clear separation between real and fake news samples. Even with the more challenging FineFake dataset, which contains greater sample diversity and more complex image-text relationships, the model maintains effective clustering performance, with feature points forming distinct elongated cluster distributions. In contrast, comparative methods (e.g., MTS, MSACA, MMFN, etc.) exhibit scattered and non-clustered data patterns. These results confirm that the HPT module effectively fuses multimodal features to capture text-image semantic relationships while preserving discriminative features for detection. Furthermore, the consistent performance across diverse datasets highlights LLM-MFEFND's strong generalization capability and superior discriminative power.

5. Discussion

5.1. Theoretical implications

The proposed LLM-MFEFND model represents a significant advancement in multimodal fake news detection by integrating LLM-generated content with hierarchical progressive fusion and explainability. Previous methods often focus on limited modalities/features, such as text or images, which can lead to suboptimal performance due to incomplete contextual understanding. Meanwhile, LLMs' direct application to fake news detection is hindered by issues like hallucinations and misalignment. To the best of our knowledge, our method is the first work to address this by enabling deep and progressive fusion of five distinct features (text, image, aligned multimodal features, background knowledge, and comments) with the assistance of LLMs, without relying on LLMs for direct classification. Our approach captures higher-order complementary information and cross-modal interactions, setting a new benchmark for multimodal fusion techniques in fake news detection. Moreover, unlike previous studies, our MLIME method provides post-hoc explanations for model decisions under multimodal scenarios.

5.2. Practical implications

The LLM-MFEFND model can offer significant real-world applications for combating fake news, with clear benefits for social media platforms, fact-checking organizations, and policymakers. The model's high accuracy (94.5% on WeiBo21, 81.1% on FineFake) enables platforms like Twitter, Facebook, and WeiBo to automatically flag suspicious content with greater reliability, reducing reliance on manual review. Besides, LLM-MFEFND can be used in fact-checkers to prioritize potentially fake news for review, leveraging its explainable outputs to quickly identify misleading elements. Moreover, according to the explainability of our model, the model helps users understand decisions by highlighting which features contributed to a "fake" classification.

5.3. Future work

While LLM-MFEFND performs well on the tested datasets, its generalizability to other languages or niche domains (e.g., scientific misinformation) remains to be validated. Expanding the evaluation to diverse datasets and domains will be a focus of our future research. Specifically, we will utilize multilingual LLMs (Pokharel et al., 2025) to extend the model to other languages, combined with adapter-based fine-tuning to reduce computational costs. For the scientific and medical misinformation domains, we will try to integrate domain-specific knowledge graphs (e.g., BioBERT (Lee et al., 2020), SciBERT (Lobanova et al., 2023)) with the HPT module to capture technical semantics.

The integration of LLMs enhances detection performance but introduces limitations regarding latency, cost, and reliability. Inference latency, which is influenced by API calls and hardware limitations, may impact real-time usage. This issue could be resolved through local deployment or upgraded hardware. The cost of mainstream models (e.g., 0.5 RMB per million tokens) remains a barrier to large-scale deployment. To ensure output reliability, the generated content in this study is manually verified by domain experts. Future work will focus on a more systematic evaluation of the performance-cost-latency trade-offs, and explore automated filtering mechanisms to enhance the trusty of LLM-generated outputs and reduce reliance on manual verification.

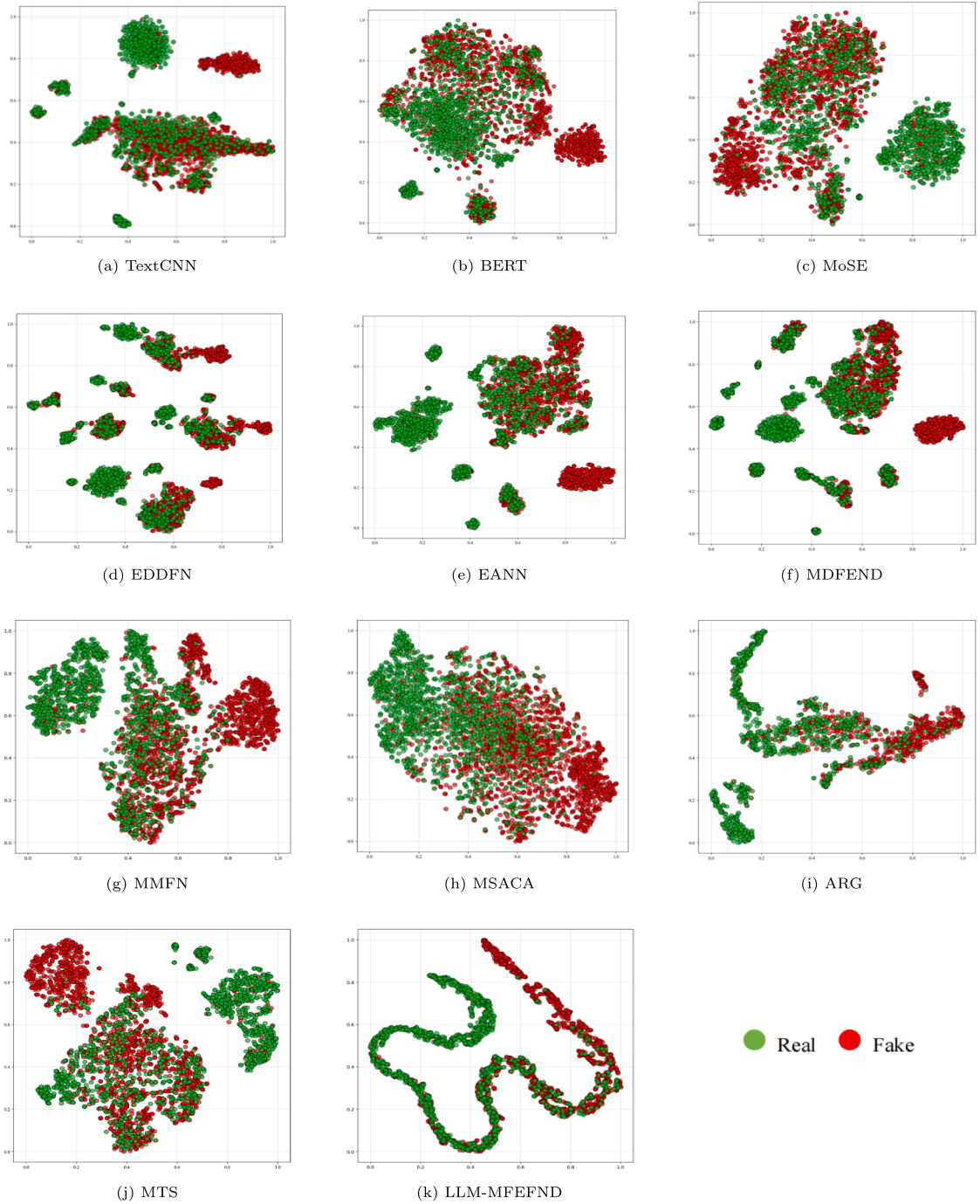


Fig. 19. Feature visualization of each representation method on FineFake.

6. Conclusion

Current news information often suffers from insufficient contextual semantics. Expanding contextual semantics while effectively integrating multi-modal features and providing interpretable analysis remains a key challenge in fake news detection. To address this issue, we propose a novel Explainable Fake News Detection method that integrates Multimodal Fusion with Large Language Model-generated content (LLM-MFEFND) model. Our method leverages DeepSeek/ChatGLM to generate background information and comments to solve the problem of sparse semantic representation and incomplete context understanding. Subsequently, we effectively integrate text, images, aligned multimodal features, and LLM-generated augmented features through the HPT module, enabling deep

and progressive fusion of various features, and addressing the limitations of previous methods that often focus on fewer features. Additionally, we design MLIME to offer transparent insights into its decisions under the situation of multimodal fusion, enhancing reliability and explainability. In the end, we further provide quantitative analysis and case studies to demonstrate the explainability of the experimental results.

We conduct experiments on the WeiBo21 and FineFake datasets, with the following key findings: (1) LLM-MFEFND-DS outperforms existing methods, achieving satisfactory and robust performance with high F1-scores (0.945 on WeiBo21, 0.807 on FineFake). Additionally, MLIME demonstrates strong explainability, showing potential for generalization to other multimodal tasks. (2) The choice of LLMs can affect fake news detection performance. Meanwhile, the effectiveness of zero-shot and few-shot strategies varies with the choice of LLMs, and few-shot does not consistently surpass zero-shot.

CRedit authorship contribution statement

Lidong Wang: Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Software, Supervision, Project administration; **Xun Li:** Software, Data curation; **Bin Zhou:** Supervision, Writing – review & editing, Project administration ; **Yin Zhang:** Visualization; **Jie Yuan:** Investigation, Validation; **Hua Hu:** Supervision, Writing – review & editing.

Data availability

The code is available on GitHub <https://github.com/li1608335419/LLM-MFEFND>, as mentioned in the experimental evaluation section.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix A. Comparison of fake news detection methods

Table A.9
Comparison of fake news detection methods.

Category	Method	Core Innovation	Features	Dataset	F1
Multi-modal methods	MCAN (Wu et al., 2021b)	Co-attention networks	Text + Image	Twitter / Weibo	0.809 / 0.899
	EMIF (Dong et al., 2024)	Multi-source fusion	Text + Context	FibVID	0.847
	ETMA (Yadav et al., 2024)	Transformer attention	Text + Image	Twitter / Jhwiki / Pontes	0.920 / 0.960 / 0.950
	MMFN (Zhou et al., 2023)	Feature fusion	Text + Image	Weibo / Twitter / GossipCop	0.924 / 0.905 / 0.684
	MSACA (Wang et al., 2024b)	Semantic alignment	Text + Image	Weibo / GossipCop	0.903 / 0.933
	KEMFND (Gao et al., 2024)	Knowledge-enhanced	Text + Image + KG	PolitiFact / GossipCop	0.919 / 0.856
	MVAEs (Gólo et al., 2023)	Variational autoencoders	Multi-modal	FCN / PNNFake.Br	0.885 / 0.396 / 0.650
	MMFEND (Tong et al., 2024)	Multi-domain detection	Text + multi-domain	Weibo21	0.9137
	BMR (Ying et al., 2023)	Multi-view representations	Text + Image	Weibo / GossipCop / Weibo21	0.914 / 0.691 / 0.927
	MMVD (Zeng et al., 2024)	Multi-view debiasing	Text + Image + Video	FakeSV / FVC	0.8264 / 0.8928
	KEAN (Zhou et al., 2024)	Domain adaptation	Text + Image	FineFake	0.797
	ARG (Hu et al., 2024)	LLM rationales	Text	GossipCop / Weibo21	0.771 / 0.778
	SELF-CHECK (Li et al., 2023)	Fact verification	Text	BINGcheck / FeverWice	0.605 / 0.430
LLM-based methods	FactAgent (Li et al., 2024)	Structured workflow	Text	PolitiFact / GossipCop / Snopes	0.830 / 0.750
	AdStyle (Park et al., 2024)	Style augmentation	Text	PolitiFact / GossipCop / Constraint	AUC:0.922 / 0.953 / 0.972
	FLAN-T5 (Kareem & Abbas, 2023)	Chain of thought	Text	LIAR	0.7741
	SFSM (Hoy & Koulouri, 2025)	Stylistic features	Text	NELA / Facebook URLs	0.620
	MRCd (Zhou et al., 2025)	Multi-round learning	Text	Pheme / Twitter16	0.814 / 0.774
	GCAN (Lu & Li, 2020)	Graph attention	Text + Graph	Twitter15 / 16	0.876 / 0.908
	DEFEND (Shu et al., 2019)	Explainable detection	Text + comments	PolitiFact / GossipCop	0.928 / 0.755
	PCAN (Khan et al., 2024)	Pyramidal attention	Text + Network	PolitiFact / GossipCop	0.951 / 0.791
	KGf (Ribeiro et al., 2016)	Knowledge graph	Text + KG + GCN	Celebrity / PolitiFact / GossipCop	0.721 / 0.853 / 0.723
	Prop-2Vec (Silva et al., 2021a)	Propagation paths	Text + Cascades	PolitiFact / GossipCop	0.928 / 0.755
Explainable methods	QA-AXDS (Chi & Liao, 2022)	Argumentation system	Text	Twitter17 / 19 / Reddit	0.741 / 0.723 / 0.926
	Multifaceted (Han et al., 2024b)	Graph reasoning	Text + KG + Graph	PolitiFact / GossipCop / Pheme	0.939 / 0.894 / 0.890
	DCCMA (Wei et al., 2025)	Cross-modal mining	Text + Image	Weibo / PHEME / Gossipcop	0.929 / 0.932 / 0.911

Appendix B. LLM comparison experiments

To comprehensively evaluate the impact of open-source LLMs on detection results, we employ Llama 2-7B, Qwen-7B, and Mistral-7B to generate augmented information while keeping the detection model architecture and training parameters constant. We re-evaluate these configurations on WeiBo21 and FineFake (see Table B.10). The results demonstrate that substituting different LLMs has a minimal impact on overall detection performance. For instance, accuracy ranges from 0.931 to 0.938 on WeiBo21 and from 0.792 to 0.794 on FineFake. Overall, ChatGLM and DeepSeek outperform other models, with DeepSeek yielding the best performance.

Furthermore, to compare GLM-4-Air and DeepSeek-v3 with frontier model (e.g., GPT-5), we select 200 samples from Chinese and English databases, respectively. For each sample, we generate backgrounds and comments using GPT-5. News professionals then evaluate the content relevance of the generated material. Overall, the content produced by GPT-5 and ChatGLM is found to have lower relevance to the original news compared to that generated by DeepSeek. The detailed comparison data is publicly available at: <https://github.com/li1608335419/LLM-MFEFND>. Overall, as shown in Table B.11, ChatGLM-4-air and DeepSeek-v3 are significantly more cost-effective than other commercial models. DeepSeek-v3 achieves a good balance between performance and cost.

Table B.10

Comparative results of open-pource LLMs experiments. The best results are bolded.

Datasets	Method	AUC	Acc.	Fake News			Real News			Avg. F1
				Prec.	Rec.	F1	Prec.	Rec.	F1	
WeiBo21	LLM-MFEFND-Llama 2	0.978	0.938	0.924	0.957	0.940	0.955	0.919	0.937	0.939
	LLM-MFEFND-Qwen	0.977	0.931	0.937	0.928	0.932	0.927	0.936	0.931	0.932
	LLM-MFEFND-Mistral	0.983	0.937	0.952	0.921	0.937	0.922	0.953	0.936	0.937
	LLM-MFEFND-GLM	0.982	0.943	0.930	0.955	0.942	0.956	0.931	0.943	0.943
	LLM-MFEFND-DS	0.983	0.945	0.945	0.942	0.944	0.945	0.948	0.946	0.945
FineFake	LLM-MFEFND-Llama 2	0.877	0.792	0.785	0.726	0.754	0.797	0.844	0.820	0.787
	LLM-MFEFND-Qwen	0.878	0.794	0.816	0.811	0.813	0.768	0.774	0.771	0.792
	LLM-MFEFND-Mistral	0.882	0.794	0.833	0.791	0.812	0.750	0.798	0.773	0.793
	LLM-MFEFND-GLM	0.881	0.800	0.797	0.730	0.762	0.801	0.854	0.827	0.795
	LLM-MFEFND-DS	0.882	0.811	0.811	0.745	0.776	0.812	0.864	0.837	0.807

Table B.11

Comparative analysis of different commercial LLMs: Context window and price.

LLM	Context Window	Input Price (per 1M Tokens)	Output Price (per 1M Tokens)
DeepSeek-v3	128K	\$0.29	\$0.43
ChatGLM-4-air	128K	\$0.07	\$0.04
GPT-5.2	400K	\$1.75	\$14.00
GPT-5	400K	\$1.25	\$10.00
Qwen3-max	256K	\$1.37	\$5.50
Mistral Large 3	256K	\$0.50	\$1.50
Llama 3.1-405b	128K	\$5.32	\$16.0

Appendix C. Efficiency comparison

In Fig. C.20, we further present a visual comparison of model scale and computational efficiency across different methods under the same hardware (NVIDIA RTX 4090, AMD EPYC 7T83) and the same software (PyTorch 2.2.1, CUDA 12.2). Our model achieves a training speed of 6.94 ms/sample and an inference speed of 6.27 ms/sample for Chinese tasks. For English tasks, the inference speed reaches 5.39 ms/second.

The parameter scale of LLM-MFEFND is comparable to MMFN but slightly exceeds MTS and MDFEND, primarily due to the integration of five modalities and the stacked HPT architecture. In terms of training and inference latency, LLM-MFEFND performs similarly to MMFN; while slightly slower than the other two baselines, it remains fully capable of meeting real-time inference requirements.

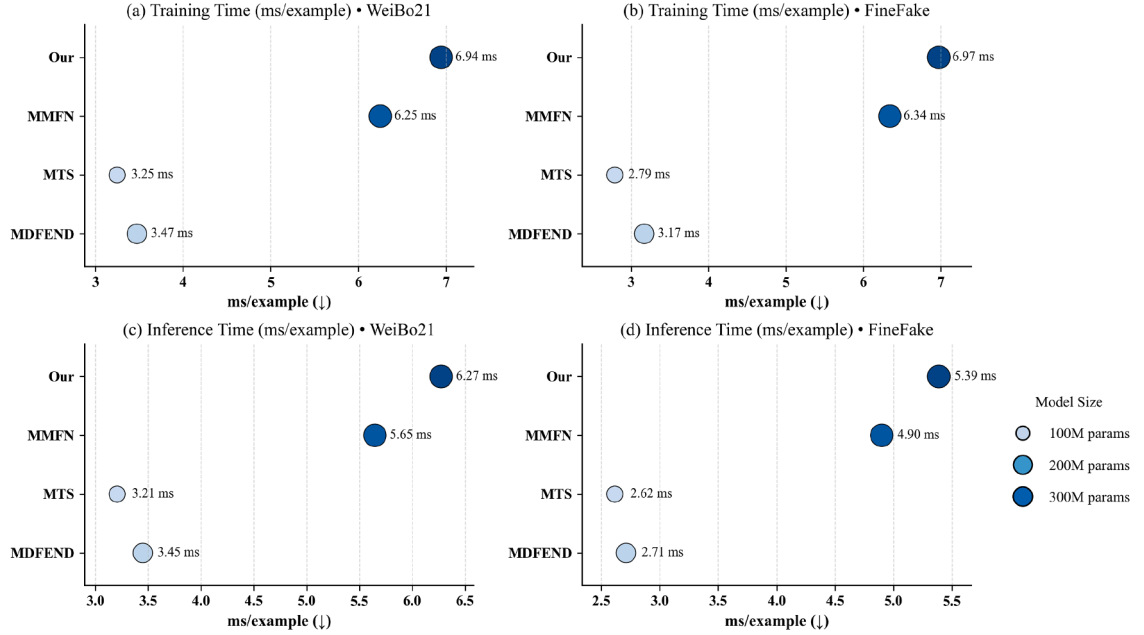


Fig. C.20. Efficiency comparison with baselines.

Appendix D. Robustness testing on LLM-generated fake news

To evaluate the robustness of LLM-MFEFND against LLM-generated fake news, we conduct comparative experiments on the MiRA-GeNews dataset (Huang et al., 2024) against strong baselines (MDFEND, MMFN, and MTS). MiRA-GeNews contains 15,000 samples featuring text and images generated by GPT-4 and Midjourney V5.2. Results show that LLM-MFEFND achieves stable and state-of-the-art performance: the GLM variant reaches an accuracy of 0.963 with balanced metrics across classes, while the DS variant further improves accuracy to 0.969 (see Table D.12).

In contrast, MDFEND suffers a significant performance drop ($\text{Acc}=0.840$), likely due to the lack of explicit domain labels in MiRA-GeNews. MMFN also underperforms, suggesting that conventional vision-language fusion is insufficient for LLM-generated content. Meanwhile, MTS exhibits a notable class imbalance (Fake Recall = 0.992 vs. Real Recall = 0.892). These findings demonstrate that LLM-MFEFND possesses superior robustness in detecting LLM-generated fake news.



Fig. D.21. Case study on MiRA-GeNews.

Table D.12

Performance of different methods on LLM-generated fake news.

Model	AUC	Acc.	Fake News			Real News		
			Prec.	Rec.	F1	Prec.	Rec.	F1
MDFEND	0.907	0.840	0.868	0.802	0.834	0.816	0.878	0.846
MMFN	0.992	0.947	0.961	0.932	0.946	0.934	0.962	0.948
MTS	0.995	0.942	0.902	0.992	0.945	0.991	0.892	0.939
LLM-MFEFND-GLM	0.993	0.963	0.964	0.963	0.962	0.962	0.964	0.963
LLM-MFEFND-DS	0.992	0.969	0.972	0.966	0.969	0.966	0.972	0.969

Fig. D.21 presents two case studies from the MiRAGeNews database. These cases represent challenging examples where the news text and accompanying images are semantically consistent, and the text itself is brief, making it difficult to detect based solely on textual content. Under such conditions, models like MDFEND, MTS, and MMFN incorrectly classify them as Real (as shown in Fig. D.21). In contrast, our approach leverages an LLM to provide the model with richer contextual information, enabling correct identification in such ambiguous cases.

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